Global Food Security and Supply Chain Analysis

Cleaning and Transformation

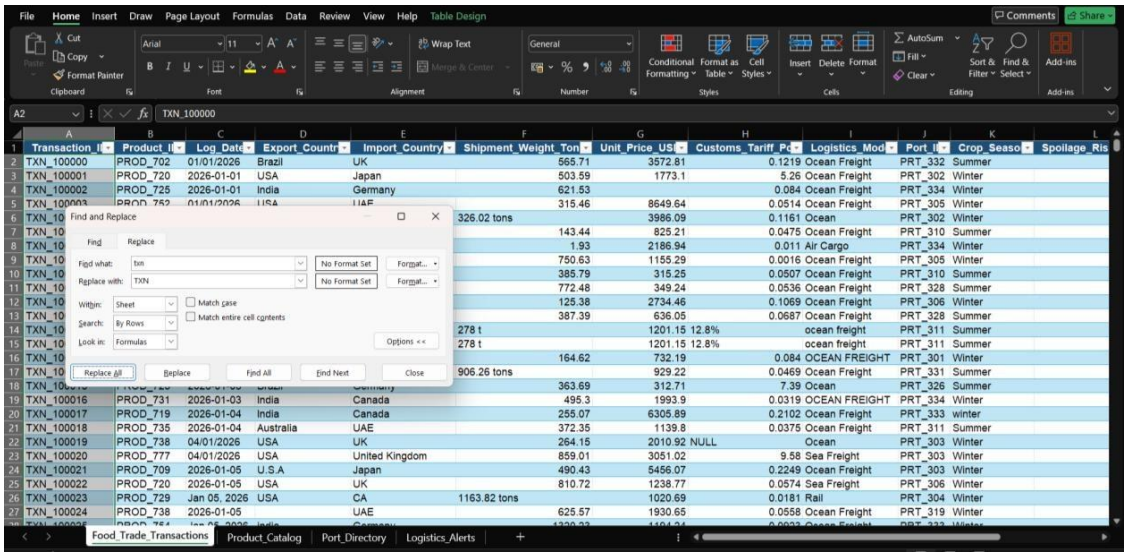
This project uses 4 tables:

- **Food_Trade_Transactions**
- **Product_Catalog**
- **Port_Directory**
- **Logistics_Alerts**.

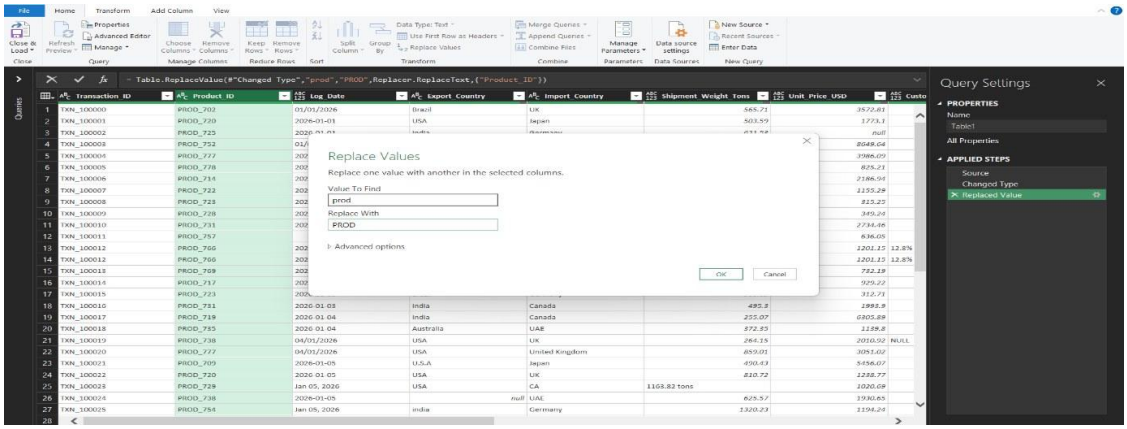
1. Food_Trade_Transactions

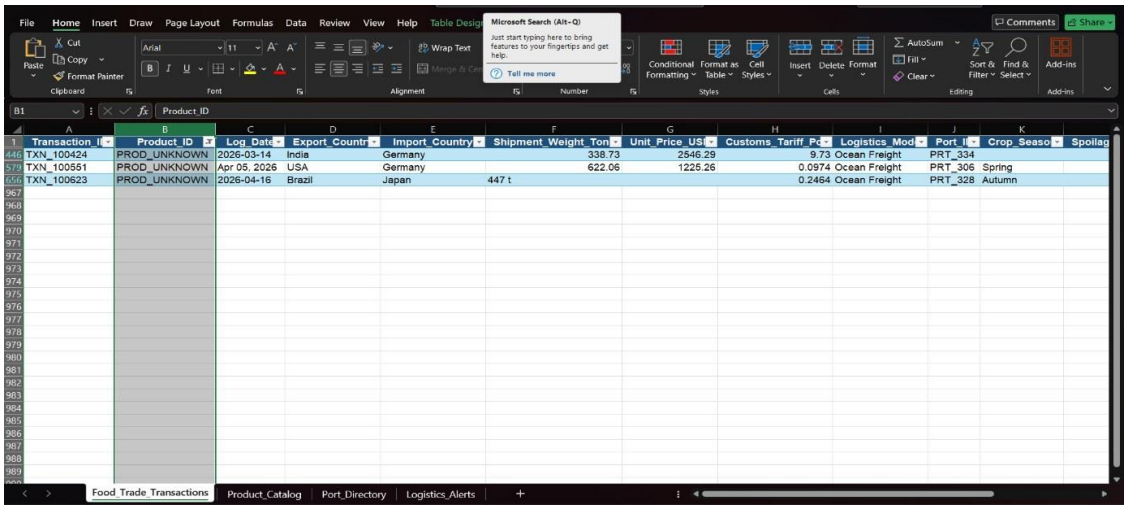
After cleaning and transformation, this table has 890 rows and 14 columns.

1. **TRANSACTION_ID**: “txn” was replaced with “TXN” using Find and Replace.

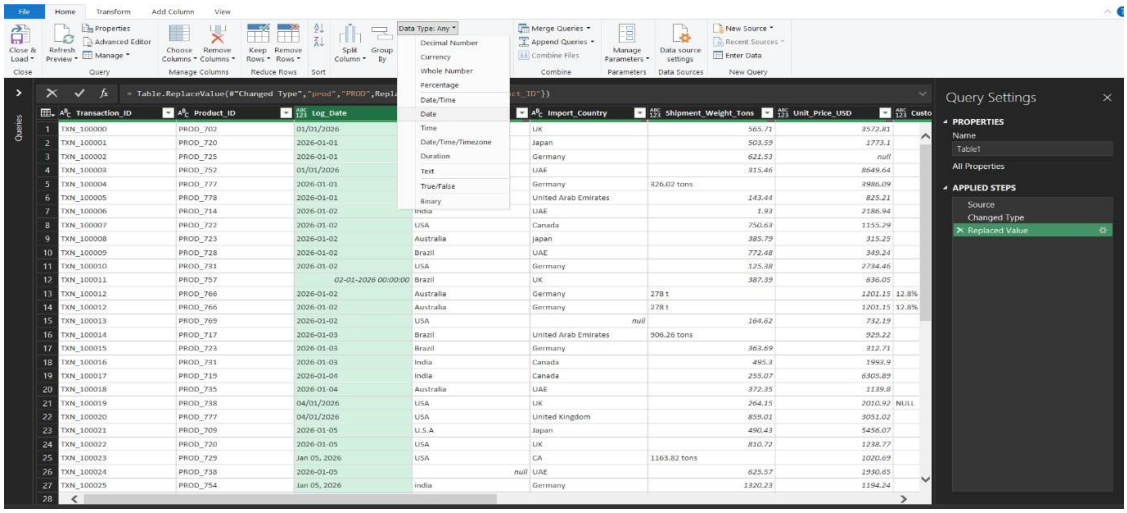


2. **PRODUCT_ID**: “prod” was replaced with “PROD” using Find and Replace, and missing values were filled with PROD_UNKNOWN.

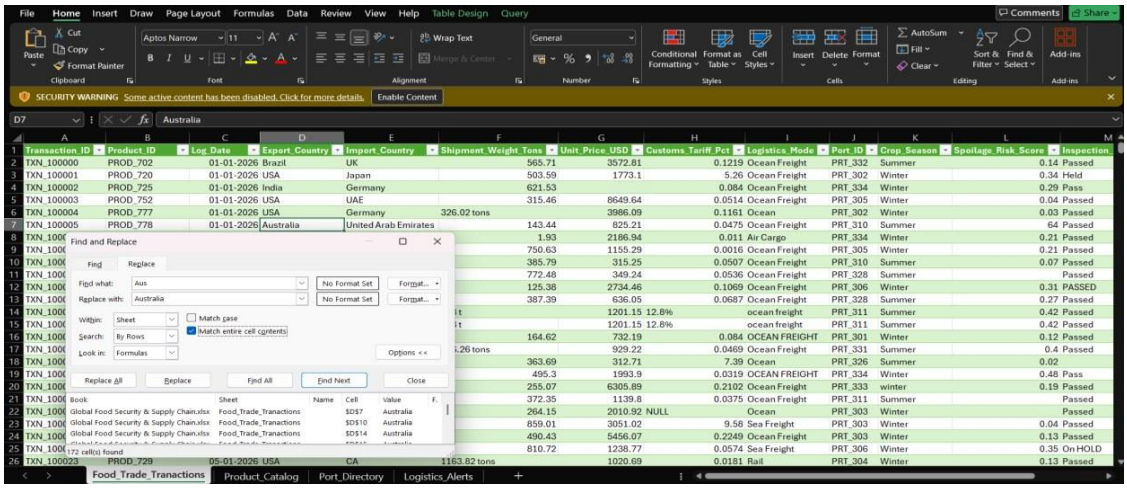


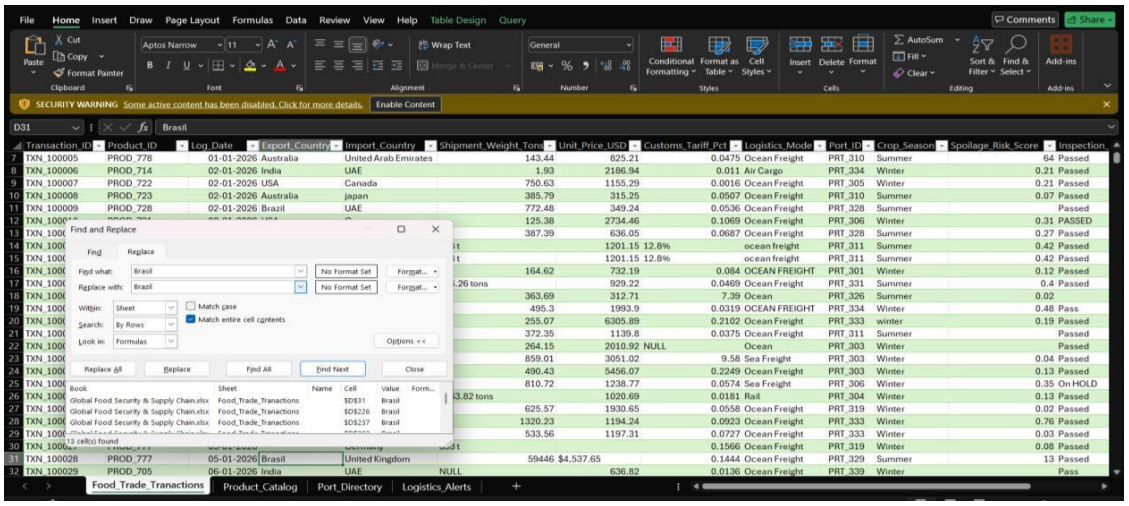


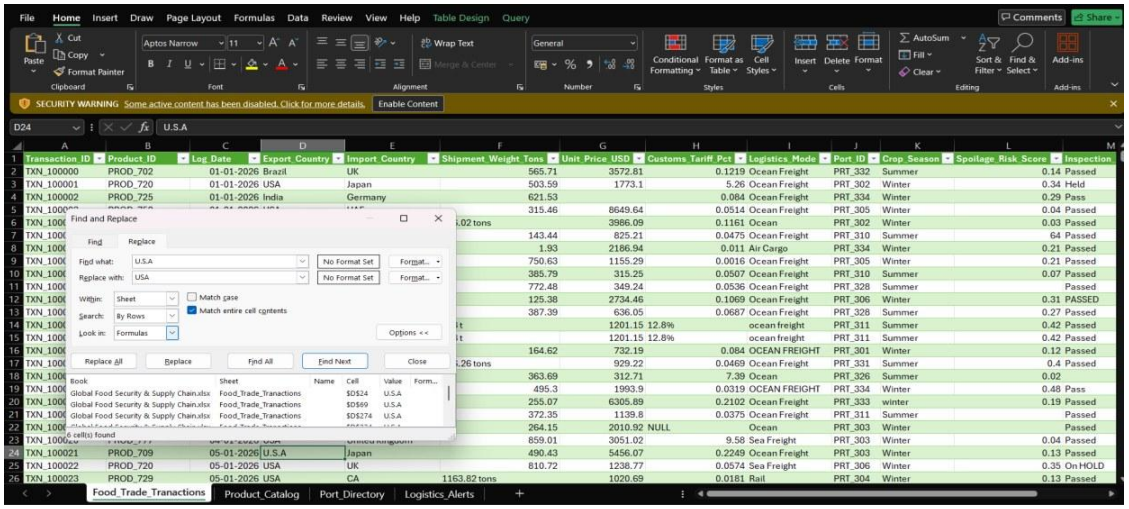
3. **LOG_DATE**: the data type was changed to Date.

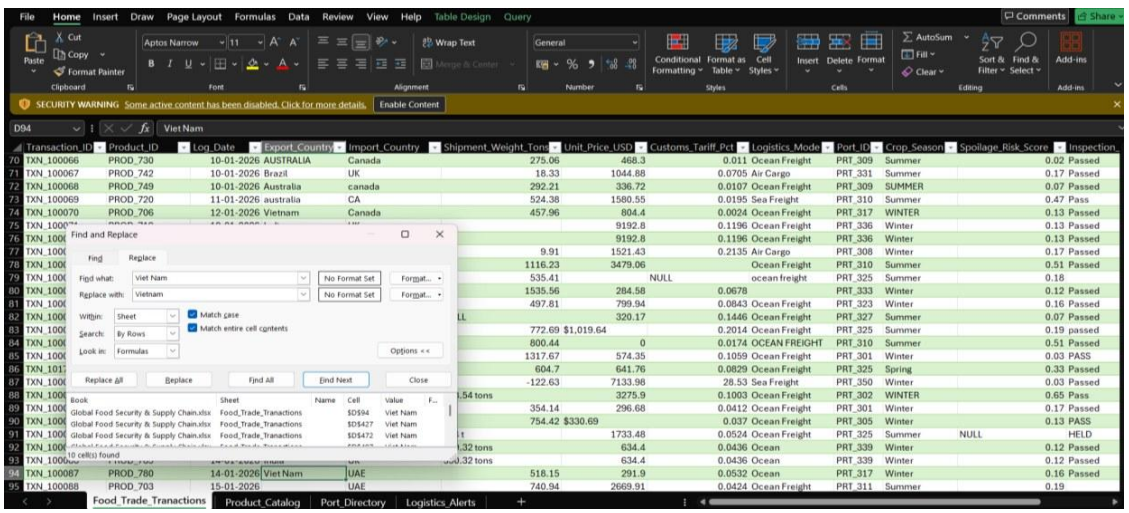


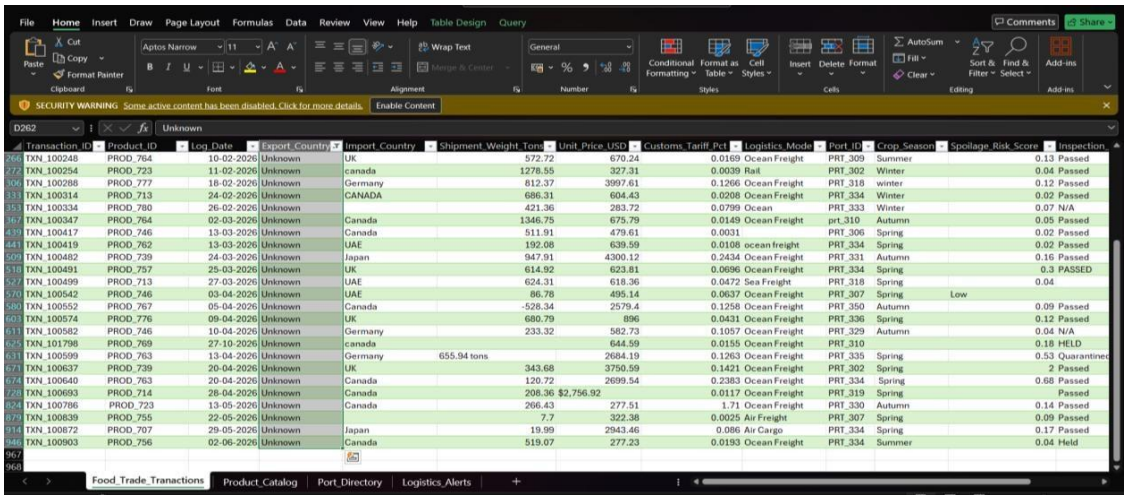
4. **EXPORT_COUNTRY**: Find and Replace was used to change “Aus” to “Australia”, “Brasil” to “Brazil”, “U.S.A” to “USA” and “Viet Nam” to “Vietnam”. Missing values were filled with Unknown.



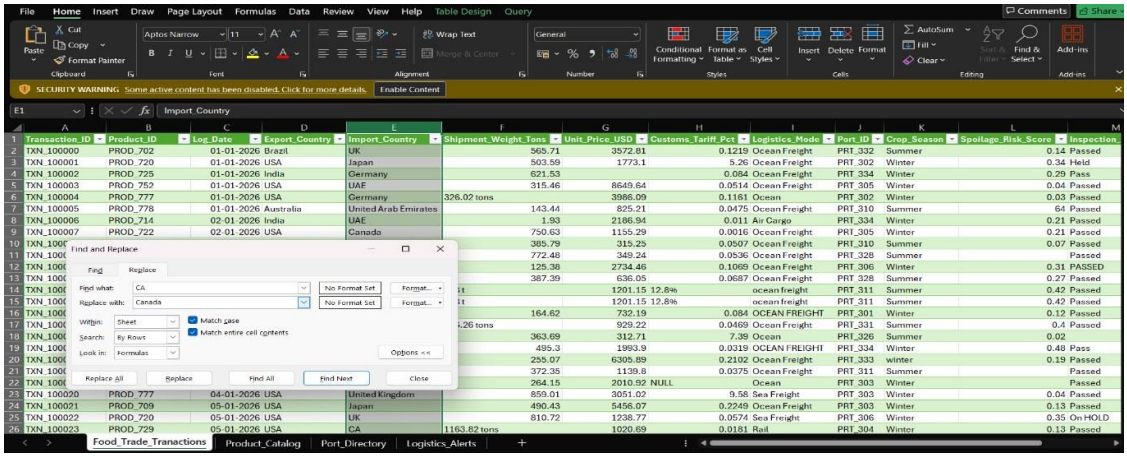

File Home Insert Draw Page Layout Formulas Data Review View Help Table Design Query
Comments Share
SECURITY WARNING Some active content has been disabled. Click for more details. Enable Content
D31 Brasil
Find and Replace
Find what: Brasil
Replace with: Brasil
No Format Set
Within: Sheet
Search: By Rows
Look in: Formulas
Match case
Match entire cell contents
Options <<
Replace All Replace Find All Find Next Close
13 cell(s) found
Food_Trade_Tranactions Product_Catalog Port_Directory Logistics_Alerts

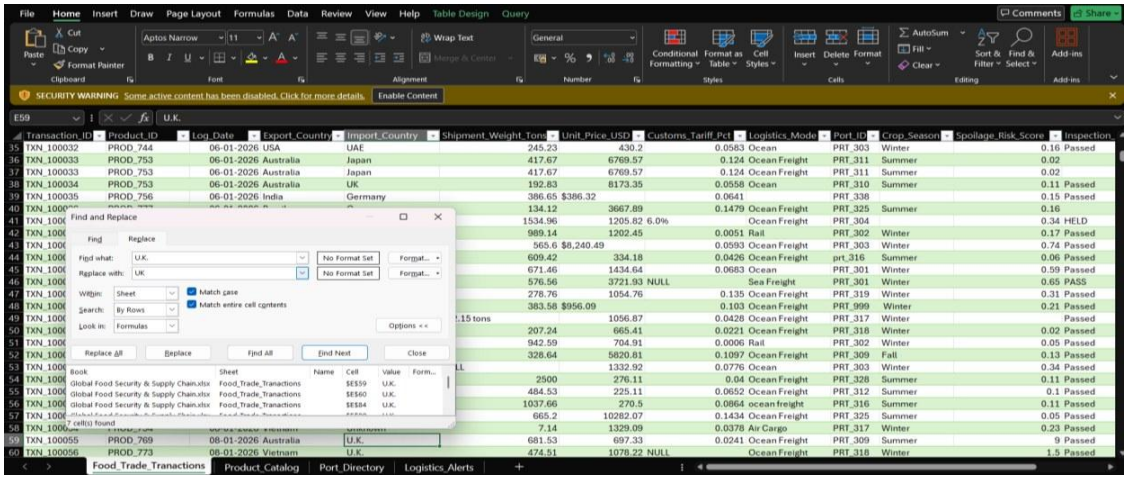

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Comments Share
SECURITY WARNING Some active content has been disabled. Click for more details. Enable Content
D24 U.S.A
Find and Replace
Find what: U.S.A
Replace with: USA
No Format Set
Within: Sheet
Search: By Rows
Look in: Formulas
Match case
Match entire cell contents
Options <<
Replace All Replace Find All Find Next Close
2 cell(s) found
Food_Trade_Tranactions Product_Catalog Port_Directory Logistics_Alerts

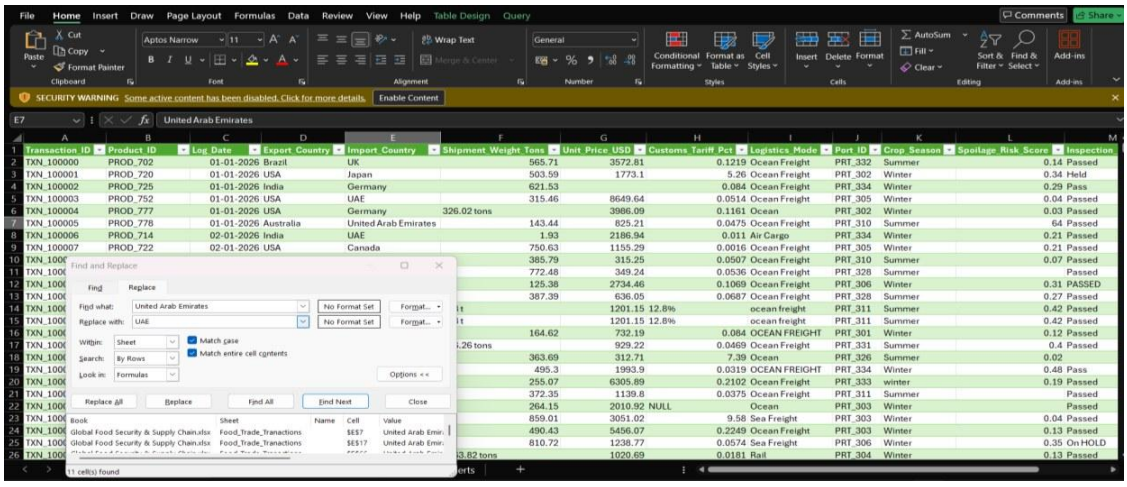

File Home Insert Draw Page Layout Formulas Data Review View Help Table Design Query
Comments Share
SECURITY WARNING Some active content has been disabled. Click for more details. Enable Content
D94 Viet Nam
Find and Replace
Find what: Viet Nam
Replace with: Vietnam
No Format Set
Within: Sheet
Search: By Rows
Look in: Formulas
Match case
Match entire cell contents
Options <<
Replace All Replace Find All Find Next Close
10 cell(s) found
Food_Trade_Tranactions Product_Catalog Port_Directory Logistics_Alerts

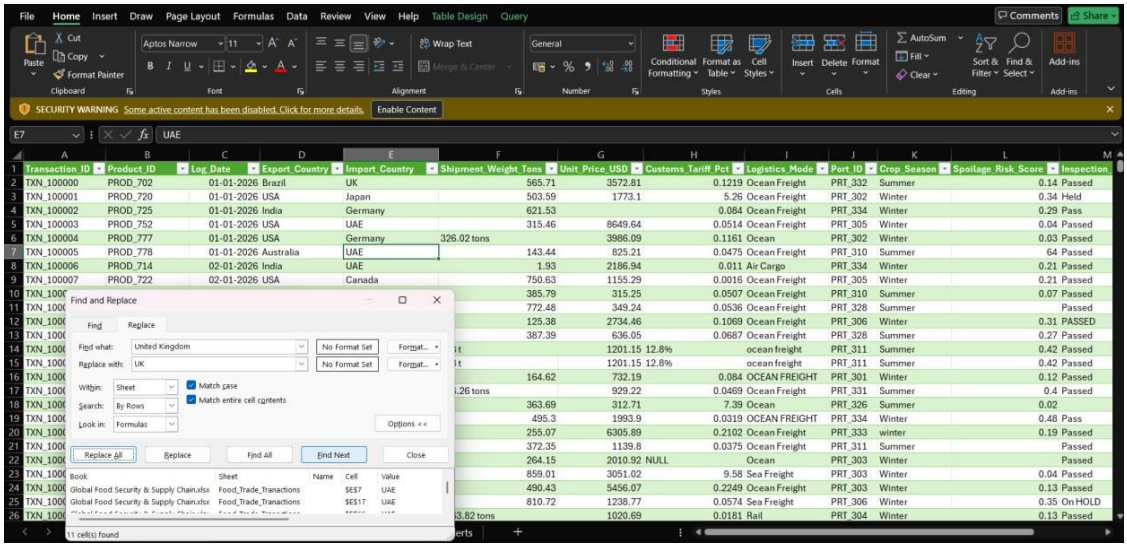


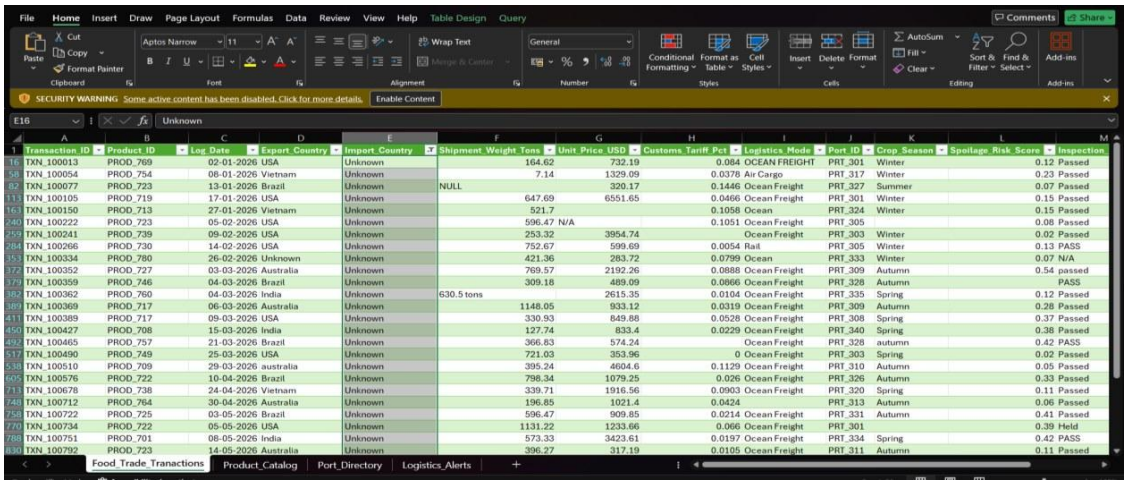
5. **IMPORT_COUNTRY**: Find and Replace was used to change “CA” to “Canada”, “U.K.” to “UK”, “United Arab Emirates” to “UAE” and “United Kingdom” to “UK”. Missing values were filled with Unknown.



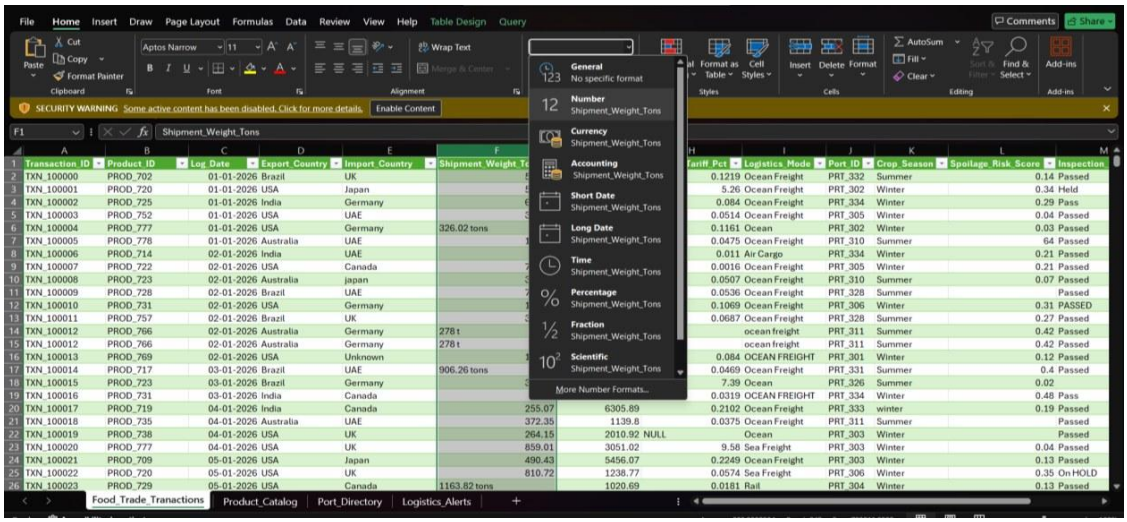


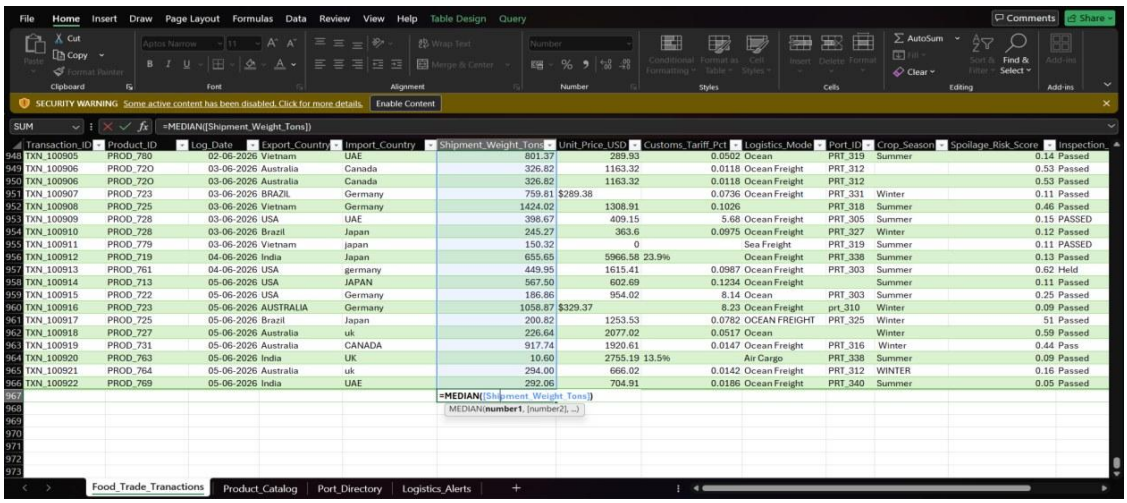

File Home Insert Draw Page Layout Formulas Data Review View Help Table Design Query
Comments Share
SECURITY WARNING Some active content has been disabled. Click for more details. Enable Content
E7 United Arab Emirates
Find and Replace
Find what: United Arab Emirates
Replace with: UAE
No Format Set
Replace All Replace Find All Find Next Close
1 cell(s) found

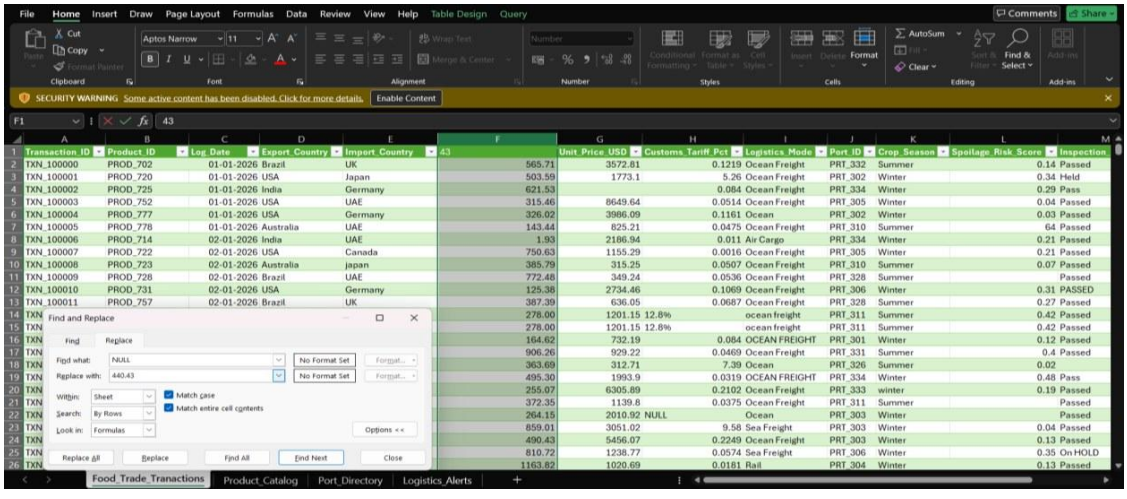

File Home Insert Draw Page Layout Formulas Data Review View Help Table Design Query
Comments Share
SECURITY WARNING Some active content has been disabled. Click for more details. Enable Content
E7 UAE
Find and Replace
Find what: United Kingdom
Replace with: UK
No Format Set
Replace All Replace Find All Find Next Close
1 cell(s) found


File Home Insert Draw Page Layout Formulas Data Review View Help Table Design Query
Comments Share
SECURITY WARNING Some active content has been disabled. Click for more details. Enable Content
E16 Unknown
Food_Trade_Tranactions
Product_Catalog
Port_Directory
Logistics_Alerts

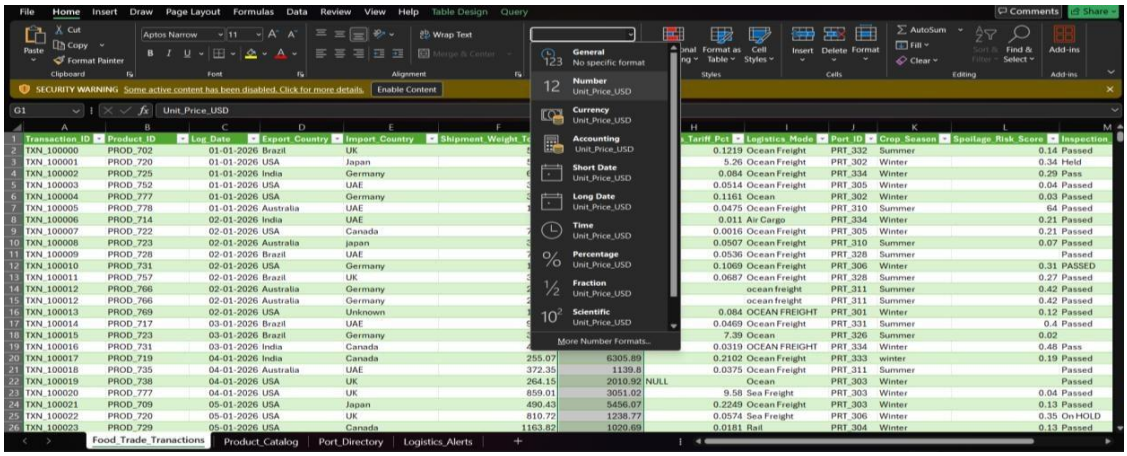
6. **SHIPMENT_WEIGHT_TONS**: the data type was changed to Number, and NULL values were filled with the median value (440.43).

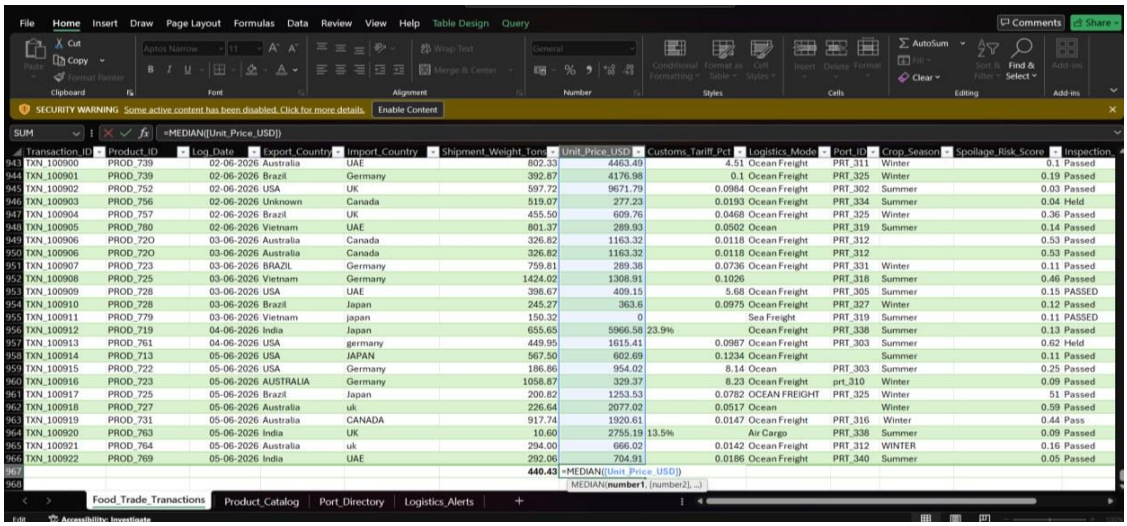




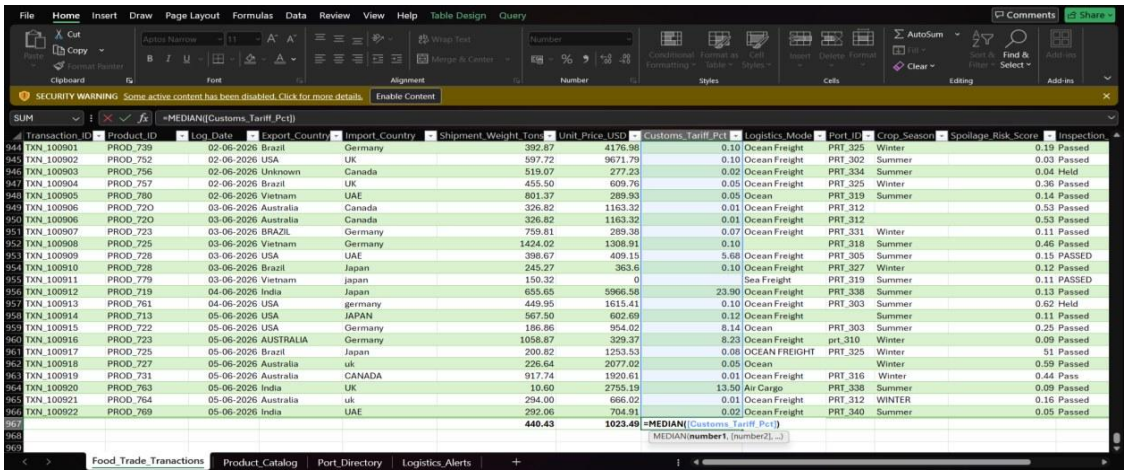


7. **UNIT_PRICE_USD**: the data type was changed to Number, and NULL values were filled with the median value (1023.49).

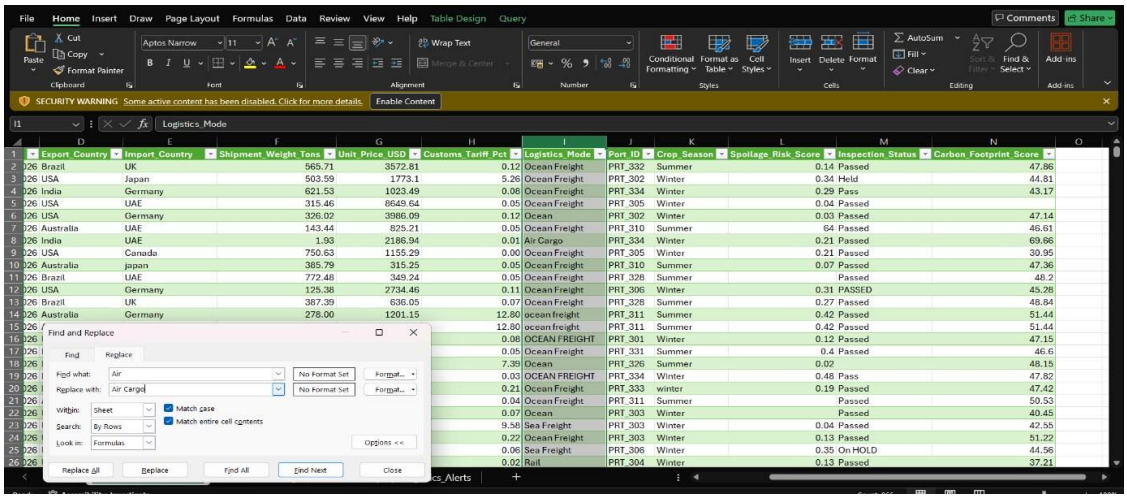


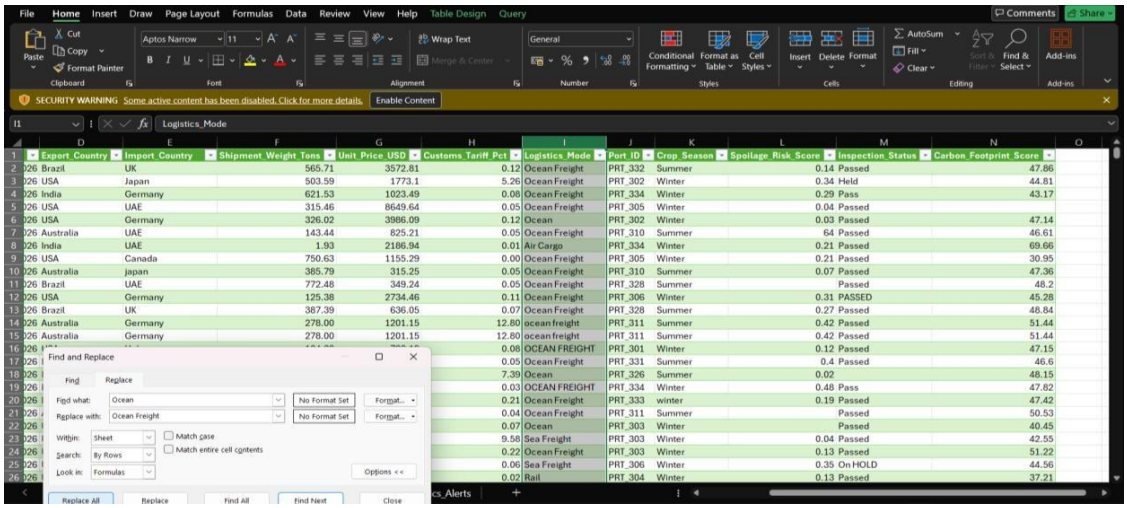


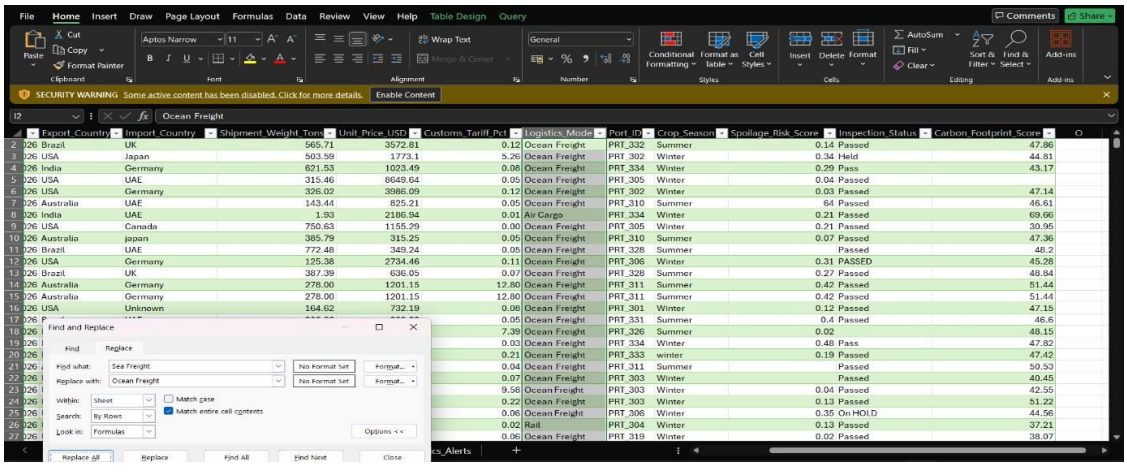
8. **CUSTOMS_TARIFF_PCT**: missing values were filled with the median value (0.07).



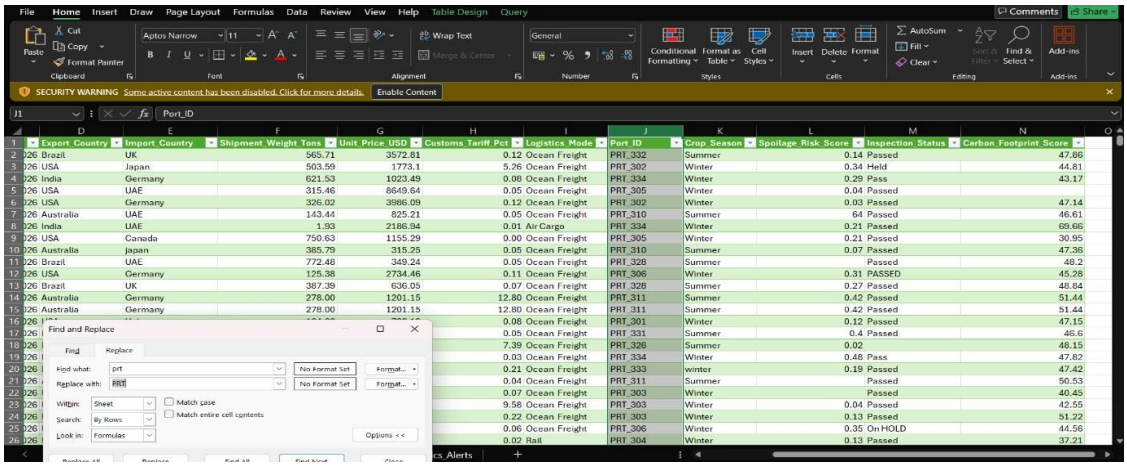
9. **LOGISTICS_MODE**: Find and Replace was used to change “Air” to “Air Cargo”, “Ocean” to “Ocean Freight” and “Sea Freight” to “Ocean Freight”.



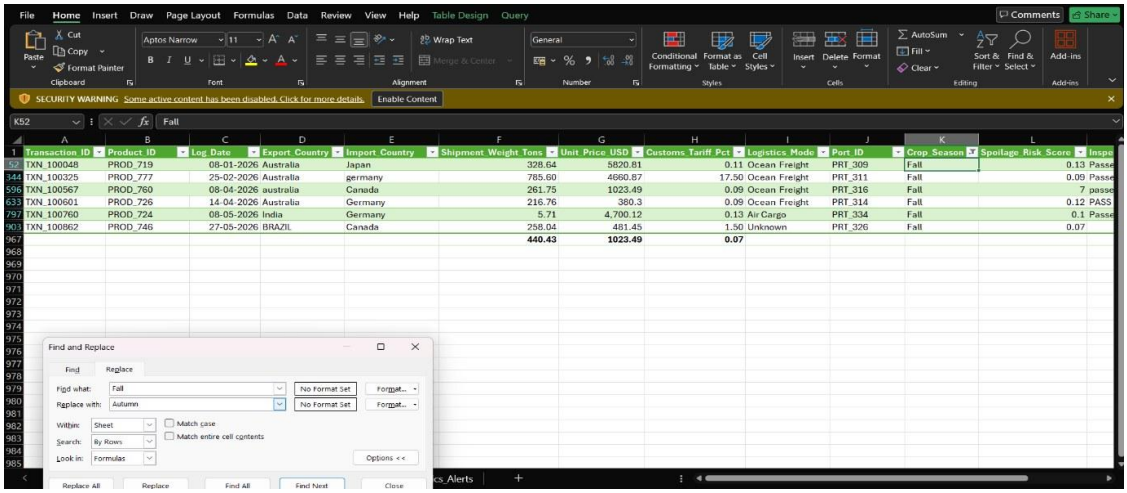


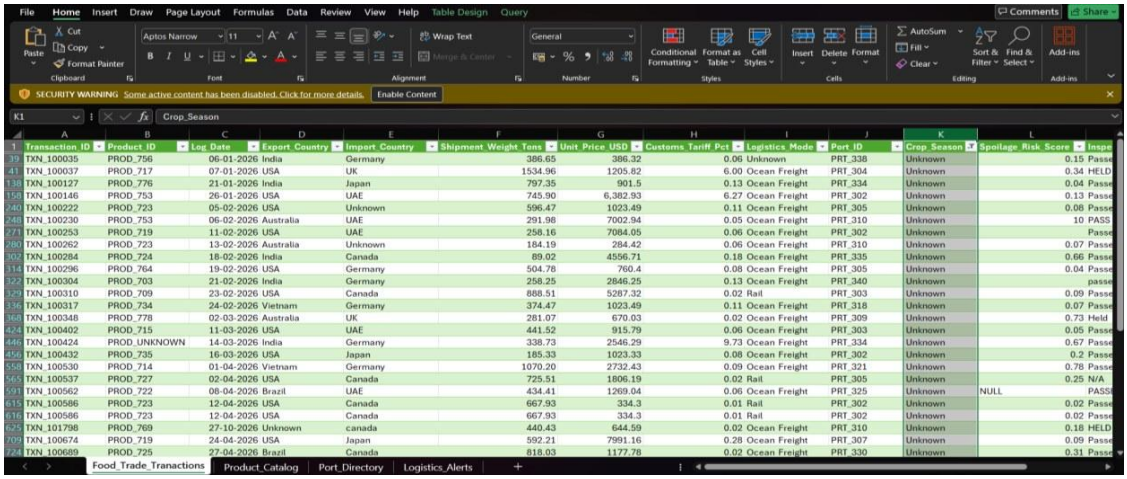


10. **PORT_ID**: “prt” was replaced with “PRT” using Find and Replace.

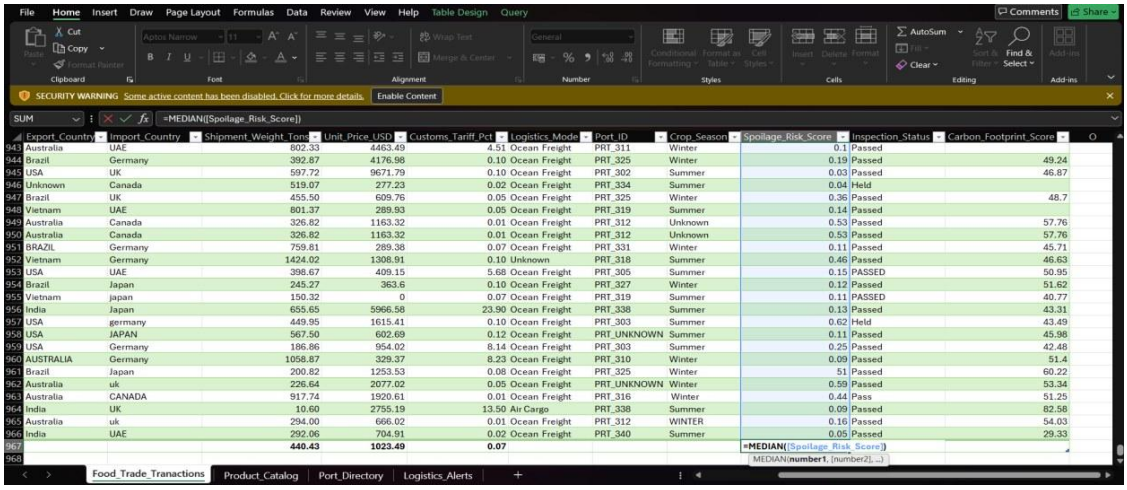


11. **CROP_SEASON**: “Fall” was replaced with “Autumn” using Find and Replace, missing values were filled with Unknown, and data inconsistencies were removed.

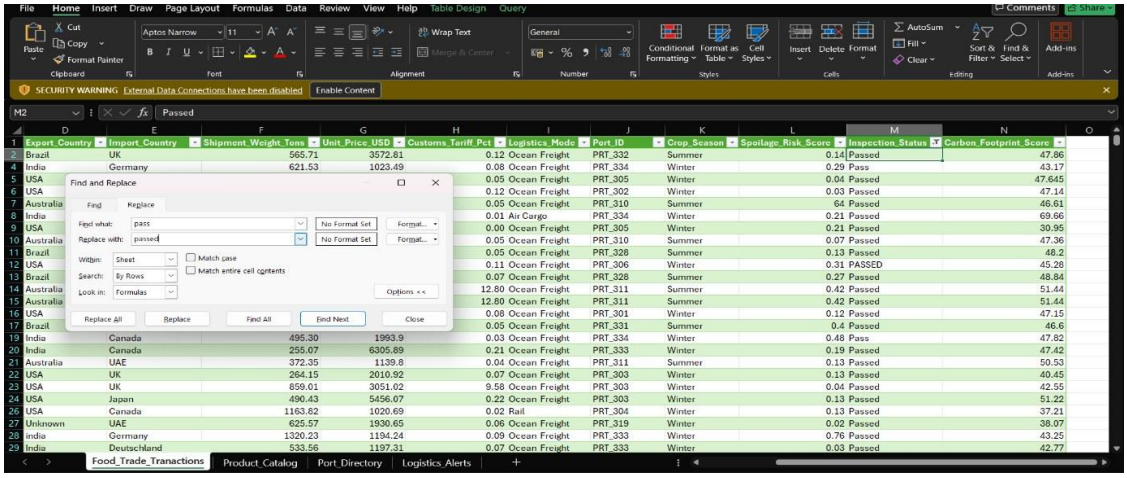


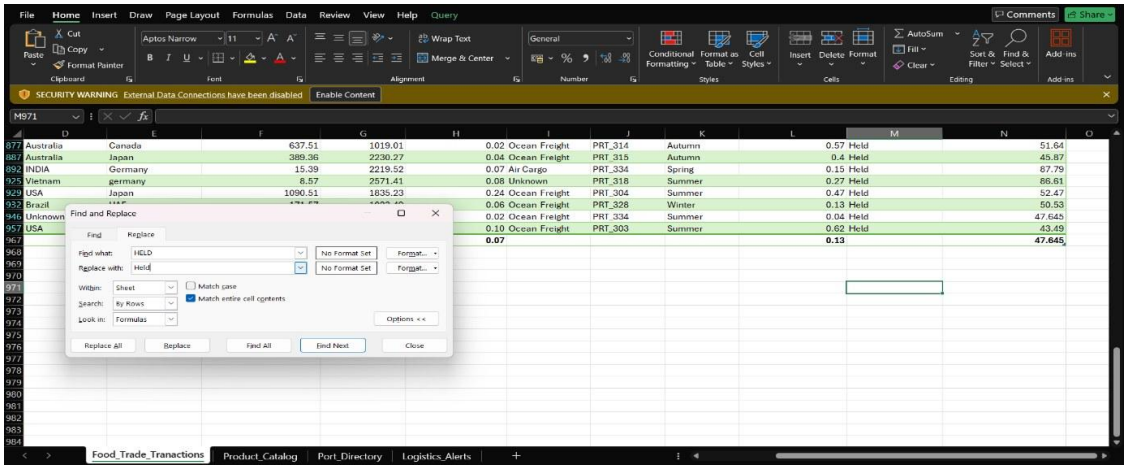


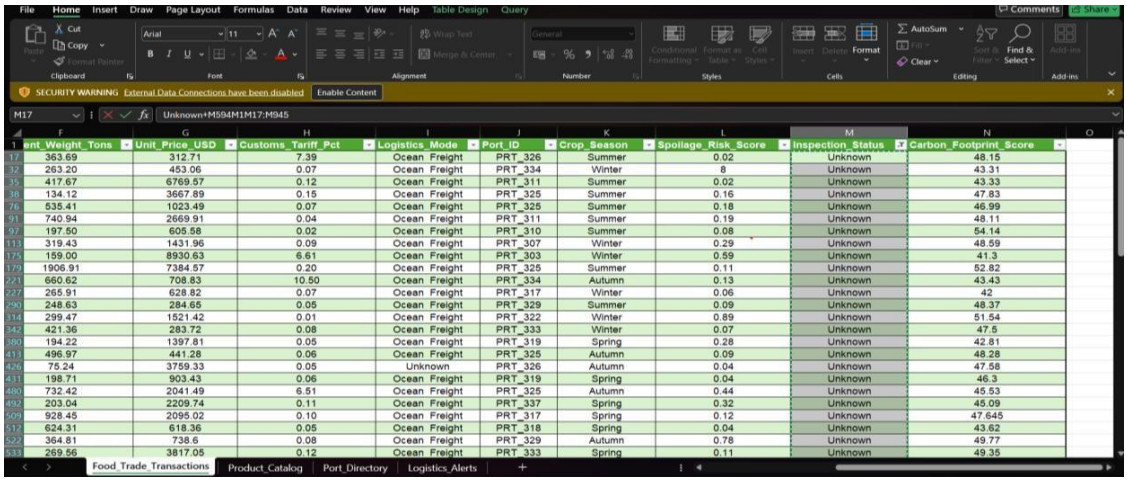
12. **SPOILAGE_RISK_SCORE**: missing values were filled with the median value (0.13).



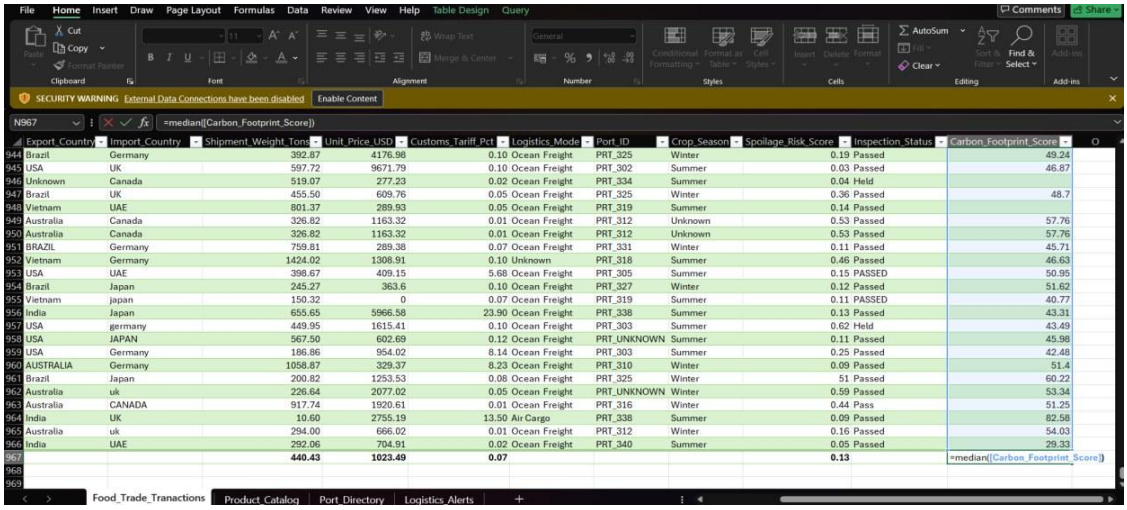
13. **INSPECTION_STATUS**: Find and Replace was used to change “pass” to “Passed” and “HELD” to “Held”, and data inconsistencies were removed.



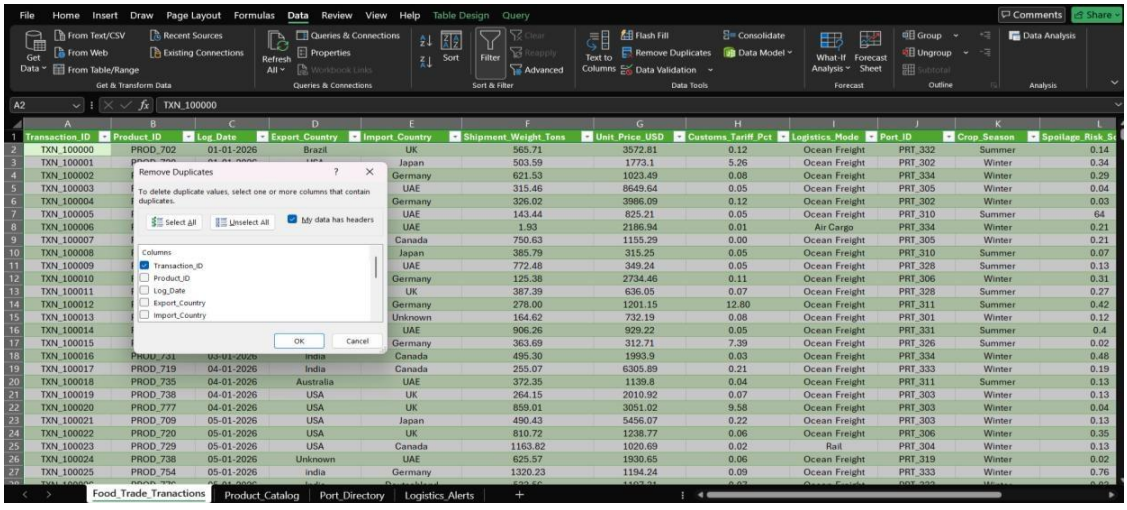


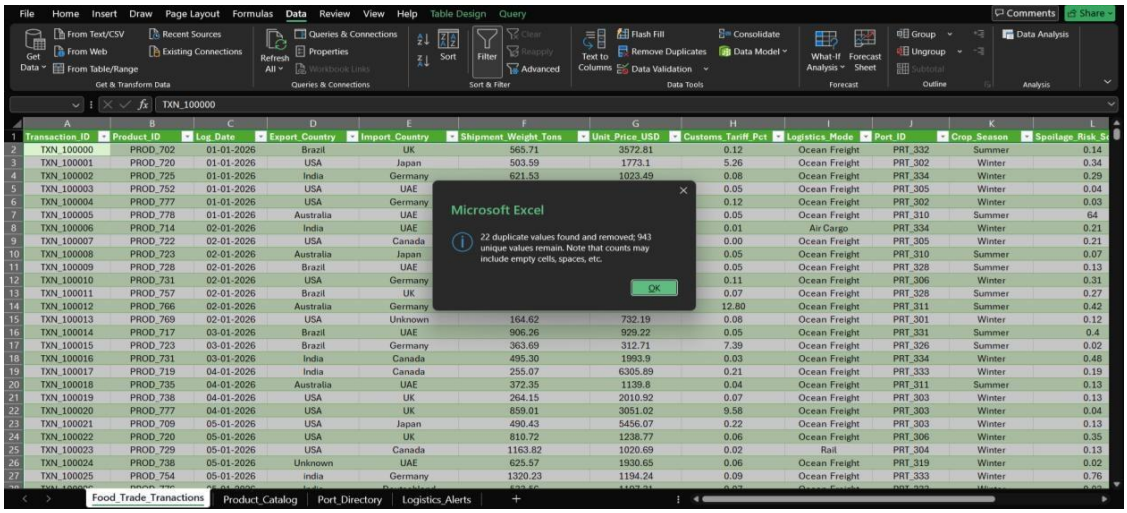


14. **CARBON_FOOTPRINT_SCORE**: missing values were filled with the median value (47.645).



Duplicate removal: 22 duplicate rows were removed using Remove Duplicates on the TRANSACTION_ID column.

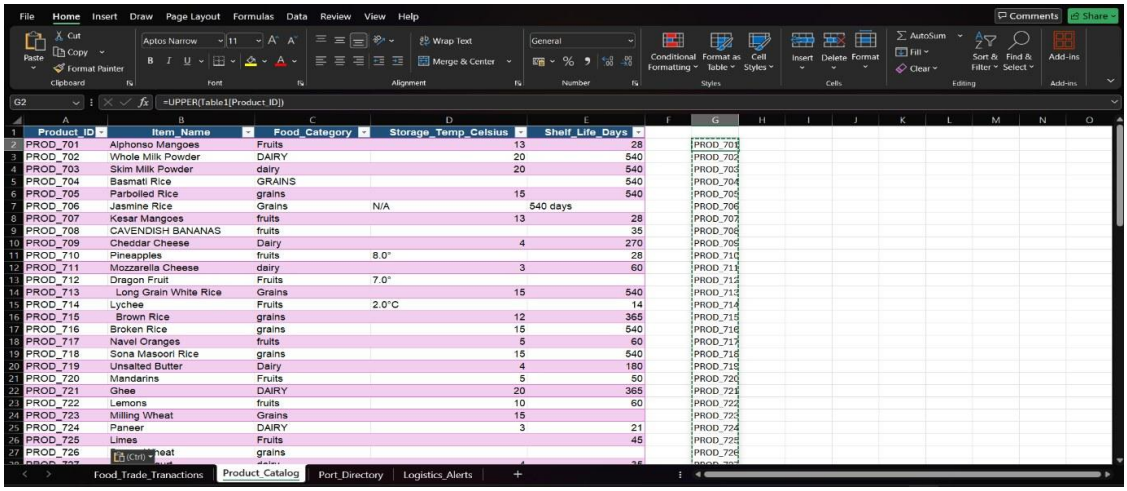




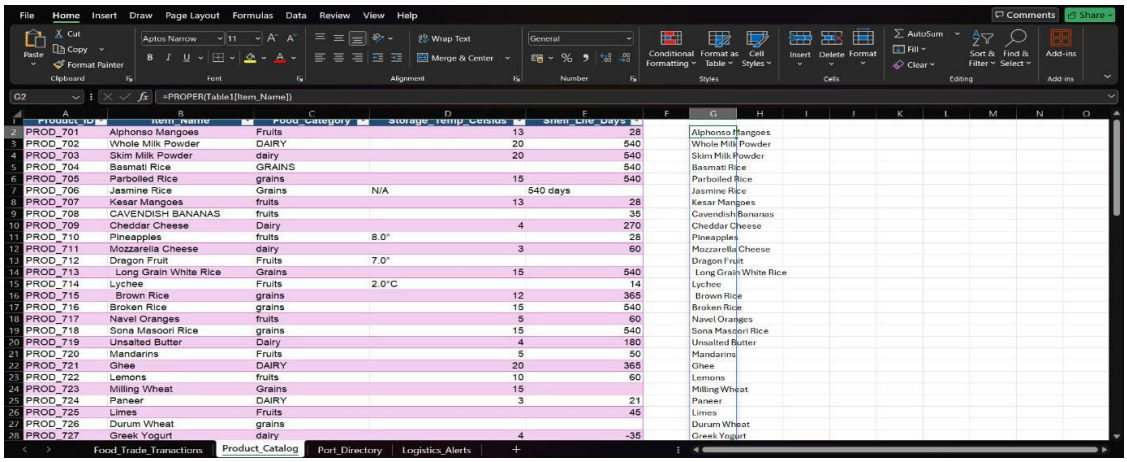
2. Product_Catalog

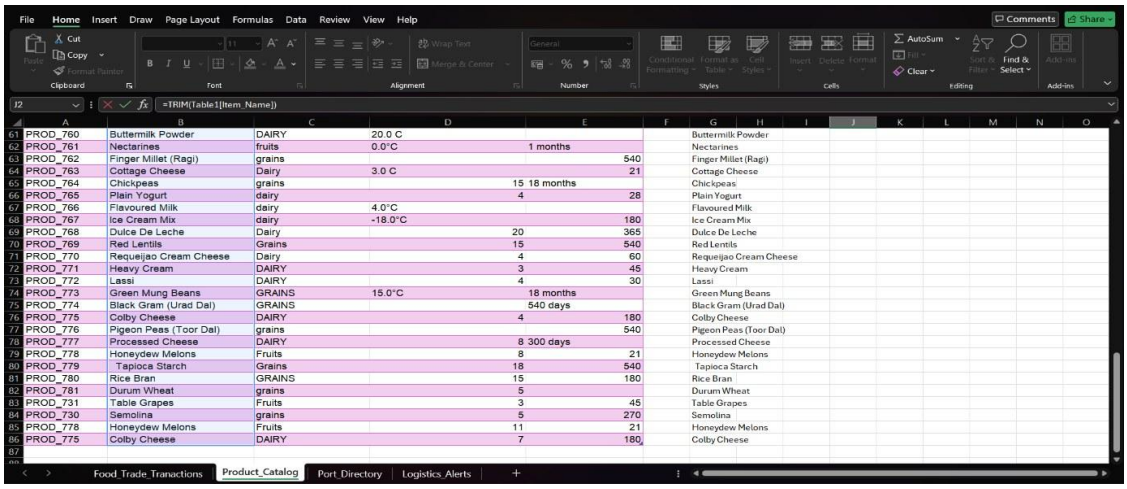
After cleaning and transformation, this table has 80 rows and 5 columns.

1. **PRODUCT_ID**: converted to upper case using the UPPER() function (for example, “prod_701” became “PROD_701”).

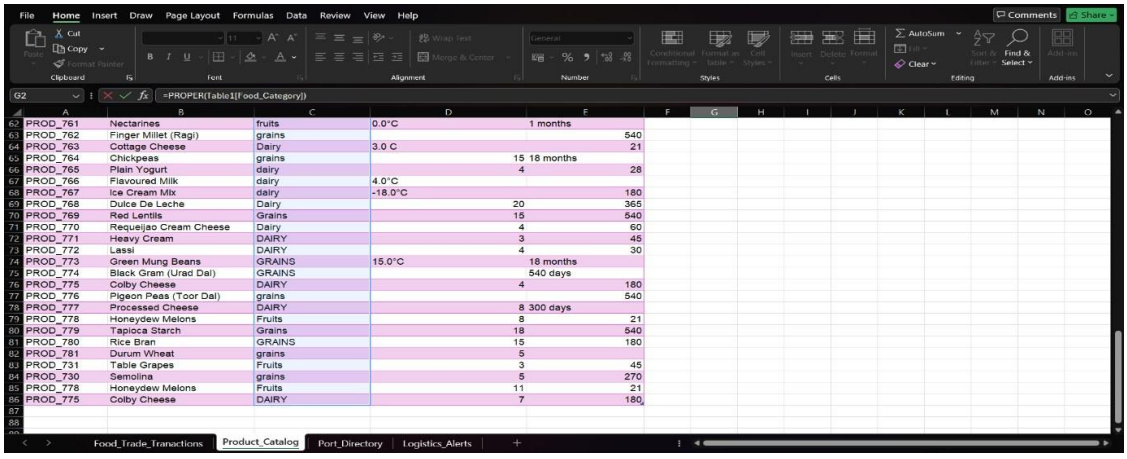


2. **ITEM_NAME**: the PROPER() and TRIM() functions were applied.

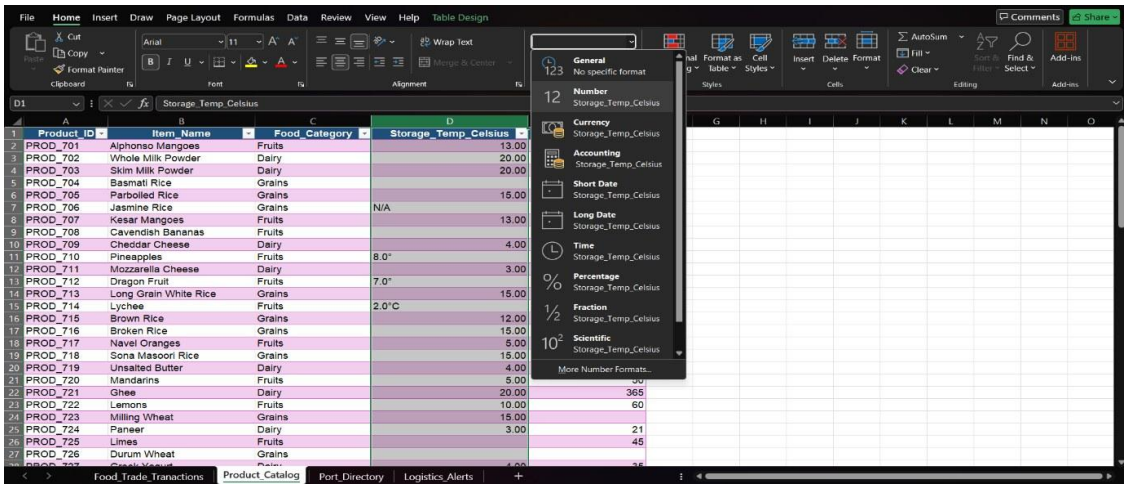


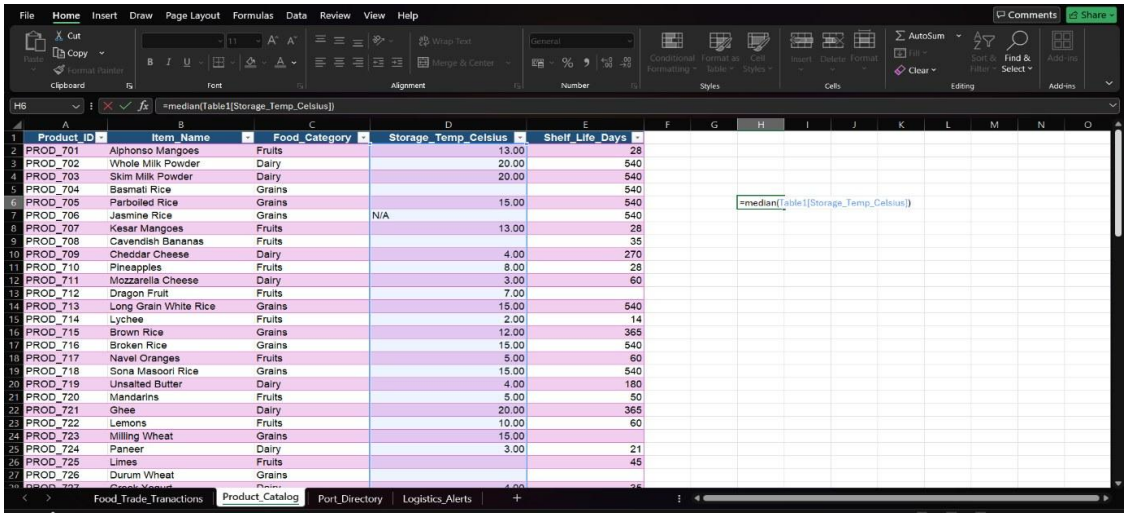


3. **FOOD_CATEGORY**: the PROPER() function was applied.

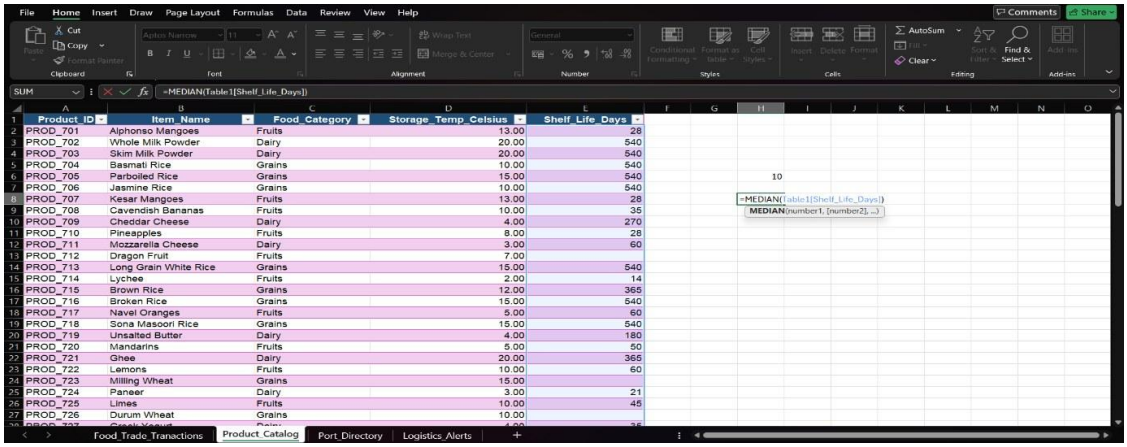


4. **STORAGE_TEMP_CELSIUS**: the data type was changed to Number, and missing values were filled with the median value (10).

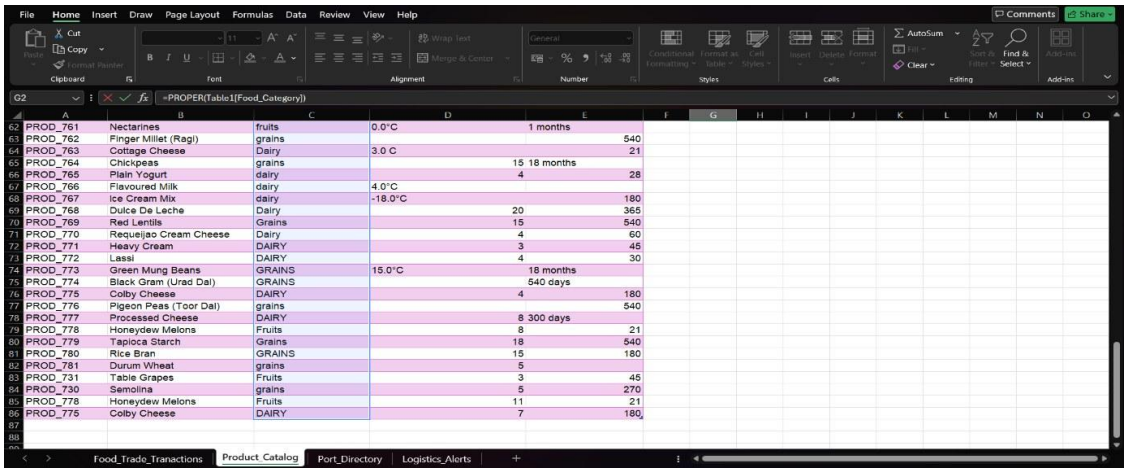




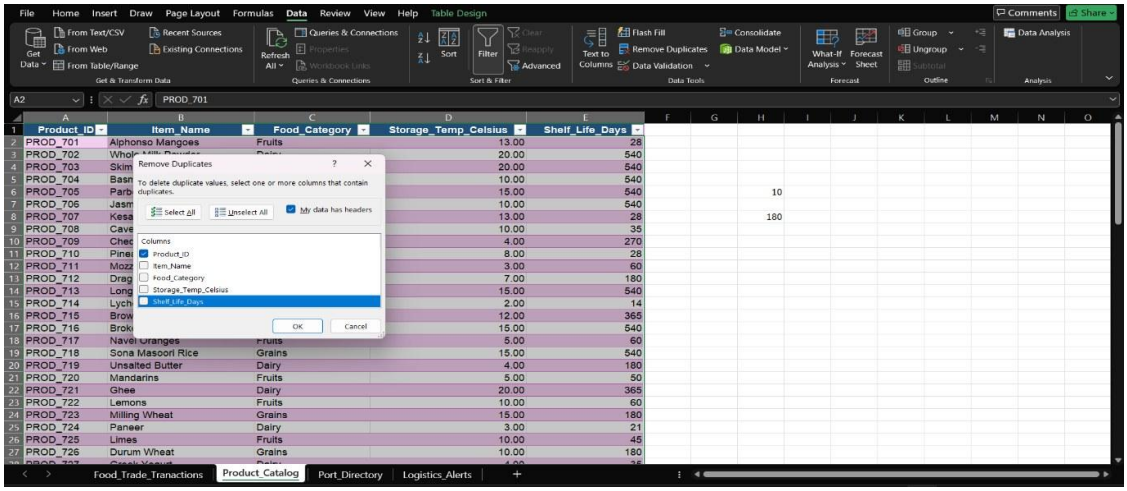
5. **SHELF_LIFE_DAYS**: the data type was changed to Number, and missing values were filled with the median value (180).

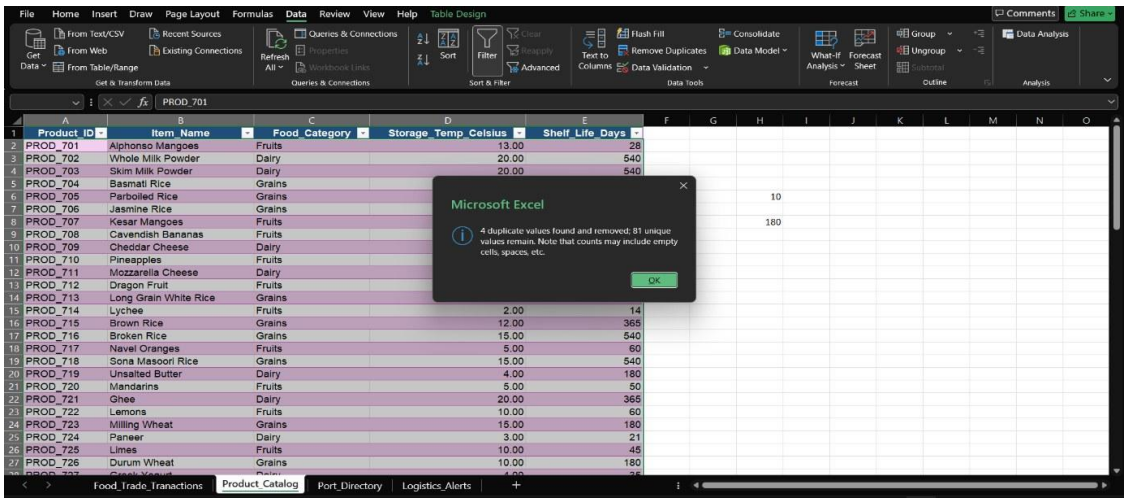


Shelf-life values given in months were converted to days
(for example, 18 months = 18 × 30 = 540 days).



Duplicate removal: 4 duplicate rows were removed using Remove Duplicates on the PRODUCT_ID column.

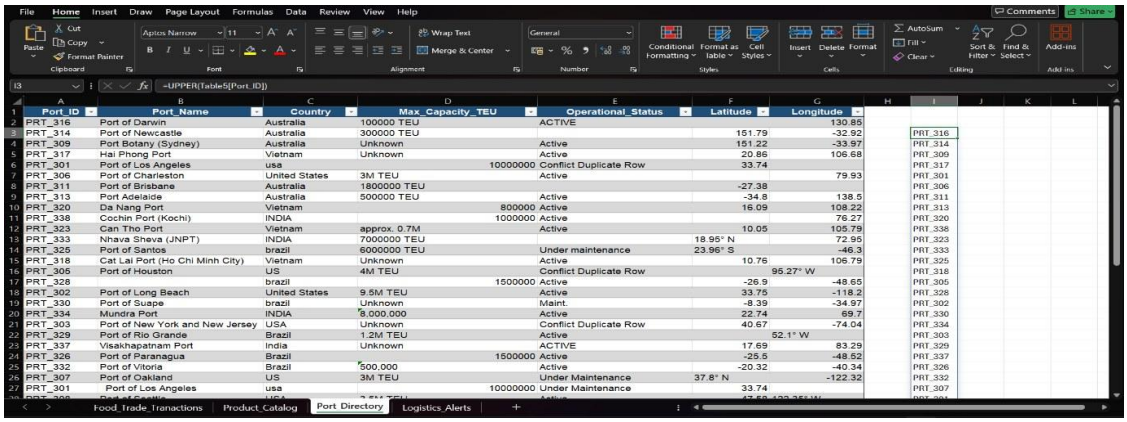




3. Port_Directory

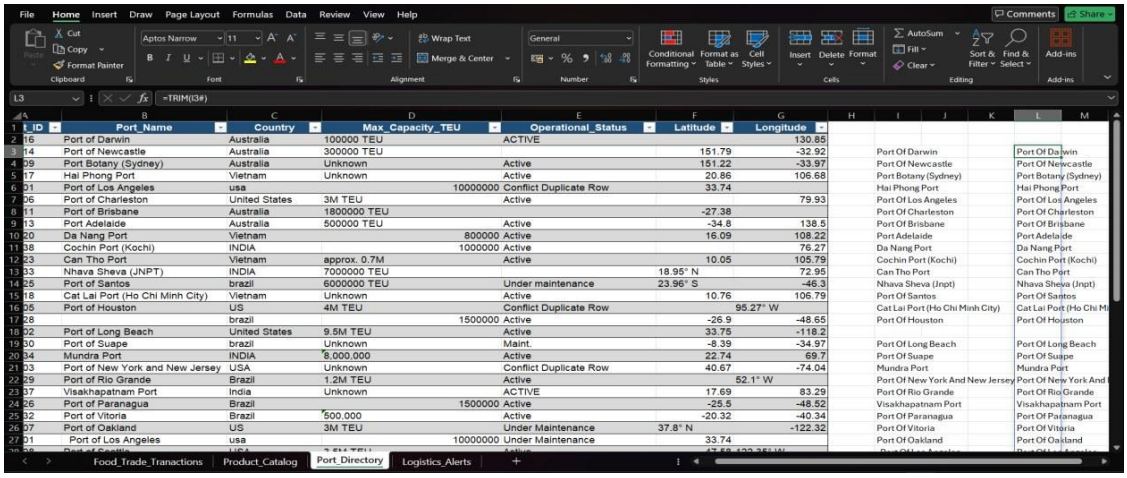
After cleaning and transformation, this table has 38 rows and 7 columns.

1. **PORT_ID**: converted to upper case using the UPPER() function (for example, “prt_314” became “PRT_314”).



Port_ID	Port_Name	Country	Max_Capacity_TEU	Operational_Status	Latitude	Longitude
PRT_316	Port of Darwin	Australia	100000 TEU	ACTIVE		130.85
PRT_314	Port of Newcastle	Australia	300000 TEU		151.79	-32.92
PRT_309	Port Botany (Sydney)	Australia	Unknown	Active	151.22	-33.97
PRT_317	Hai Phong Port	Vietnam	Unknown	Active	20.86	106.68
PRT_301	Port of Los Angeles	usa	10000000	Conflict Duplicate Row	33.74	
PRT_306	Port of Charleston	United States	3M TEU	Active		79.93
PRT_311	Port of Brisbane	Australia	1800000 TEU		-27.38	
PRT_313	Port Adelaide	Australia	500000 TEU	Active	-34.8	138.5
PRT_320	Da Nang Port	Vietnam	800000	Active	16.09	108.22
PRT_338	Cochin Port (Kochi)	INDIA	1000000	Active		76.27
PRT_323	Can Tho Port	Vietnam	approx. 0.7M	Active	10.05	105.79
PRT_333	Nhava Sheva (JNPT)	INDIA	7000000 TEU		18.95° N	72.95
PRT_325	Port of Santos	brazil	6000000 TEU	Under maintenance	23.96° S	-46.3
PRT_318	Cat Lai Port (Ho Chi Minh City)	Vietnam	Unknown	Active	10.76	106.79
PRT_305	Port of Houston	US	4M TEU	Conflict Duplicate Row		95.27° W
PRT_328		brazil	1500000	Active	-26.9	-48.65
PRT_302	Port of Long Beach	United States	9.5M TEU	Active	33.75	-118.2
PRT_330	Port of Suape	brazil	Unknown	Maint.	-8.39	-34.97
PRT_334	Mundra Port	INDIA	8,000,000	Active	22.74	69.7
PRT_303	Port of New York and New Jersey	USA	Unknown	Conflict Duplicate Row	40.67	-74.04
PRT_329	Port of Rio Grande	Brazil	1.2M TEU	Active		52.1° W
PRT_337	Visakhapatnam Port	India	Unknown	ACTIVE	17.69	83.29
PRT_326	Port of Paranagua	Brazil	1500000	Active	-25.5	-48.52
PRT_332	Port of Vitoria	Brazil	500,000	Active	-20.32	-40.34
PRT_307	Port of Oakland	US	3M TEU	Under Maintenance	37.8° N	-122.32
PRT_301	Port of Los Angeles	usa	10000000	Under Maintenance	33.74	

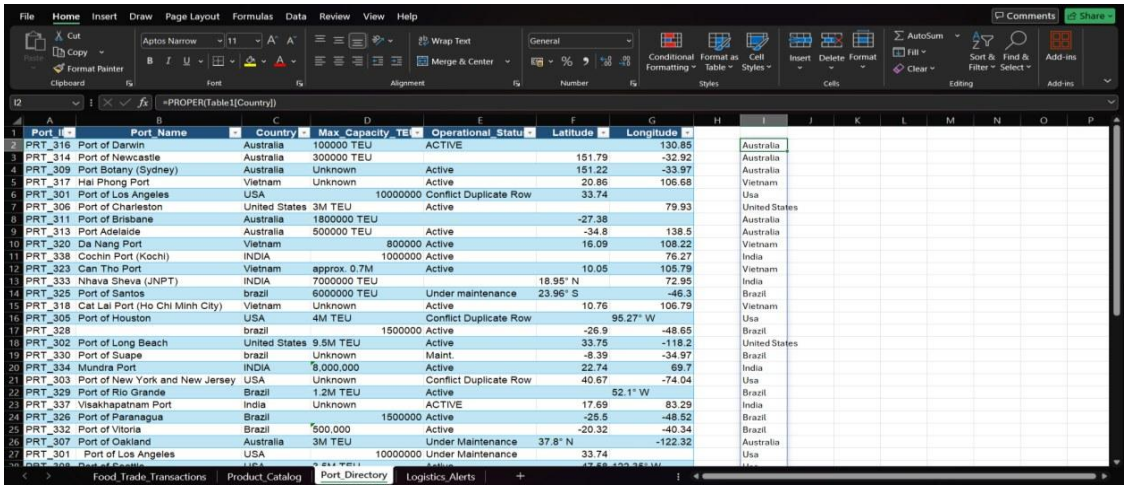
2. **PORT_NAME**: the PROPER() and TRIM() functions were applied.



Port_ID	Port_Name	Country	Max_Capacity_TEU	Operational_Status	Latitude	Longitude
16	Port of Darwin	Australia	100000 TEU	ACTIVE		130.85
14	Port of Newcastle	Australia	300000 TEU		151.79	-32.92
09	Port Botany (Sydney)	Australia	Unknown	Active	151.22	-33.97
17	Hai Phong Port	Vietnam	Unknown	Active	20.86	106.68
01	Port of Los Angeles	usa	10000000	Conflict Duplicate Row	33.74	
06	Port of Charleston	United States	3M TEU	Active		79.93
11	Port of Brisbane	Australia	1800000 TEU		-27.38	
13	Port Adelaide	Australia	500000 TEU	Active	-34.8	138.5
20	Da Nang Port	Vietnam	800000	Active	16.09	108.22
38	Cochin Port (Kochi)	INDIA	1000000	Active		76.27
23	Can Tho Port	Vietnam	approx. 0.7M	Active	10.05	105.79
33	Nhava Sheva (JNPT)	INDIA	7000000 TEU		18.95° N	72.95
25	Port of Santos	brazil	6000000 TEU	Under maintenance	23.96° S	-46.3
18	Cat Lai Port (Ho Chi Minh City)	Vietnam	Unknown	Active	10.76	106.79
05	Port of Houston	US	4M TEU	Conflict Duplicate Row		95.27° W
28		brazil	1500000	Active	-26.9	-48.65
02	Port of Long Beach	United States	9.5M TEU	Active	33.75	-118.2
30	Port of Suape	brazil	Unknown	Maint.	-8.39	-34.97
34	Mundra Port	INDIA	8,000,000	Active	22.74	69.7
03	Port of New York and New Jersey	USA	Unknown	Conflict Duplicate Row	40.67	-74.04
29	Port of Rio Grande	Brazil	1.2M TEU	Active		52.1° W
37	Visakhapatnam Port	India	Unknown	ACTIVE	17.69	83.29
26	Port of Paranagua	Brazil	1500000	Active	-25.5	-48.52
32	Port of Vitoria	Brazil	500,000	Active	-20.32	-40.34
07	Port of Oakland	US	3M TEU	Under Maintenance	37.8° N	-122.32
01	Port of Los Angeles	usa	10000000	Under Maintenance	33.74	

Port Of Darwin	Port Of Darwin
Port Of Newcastle	Port Of Newcastle
Port Botany (Sydney)	Port Botany (Sydney)
Hai Phong Port	Hai Phong Port
Port Of Los Angeles	Port Of Los Angeles
Port Of Charleston	Port Of Charleston
Port Of Brisbane	Port Of Brisbane
Port Adelaide	Port Adelaide
Da Nang Port	Da Nang Port
Cochin Port (Kochi)	Cochin Port (Kochi)
Can Tho Port	Can Tho Port
Nhava Sheva (Jnpt)	Nhava Sheva (Jnpt)
Port Of Santos	Port Of Santos
Cat Lai Port (Ho Chi Minh City)	Cat Lai Port (Ho Chi Mi
Port Of Houston	Port Of Houston
Port Of Long Beach	Port Of Long Beach
Port Of Suape	Port Of Suape
Mundra Port	Mundra Port
Port Of New York And New Jersey	Port Of New York And
Port Of Rio Grande	Port Of Rio Grande
Visakhapatnam Port	Visakhapatnam Port
Port Of Paranagua	Port Of Paranagua
Port Of Vitoria	Port Of Vitoria
Port Of Oakland	Port Of Oakland

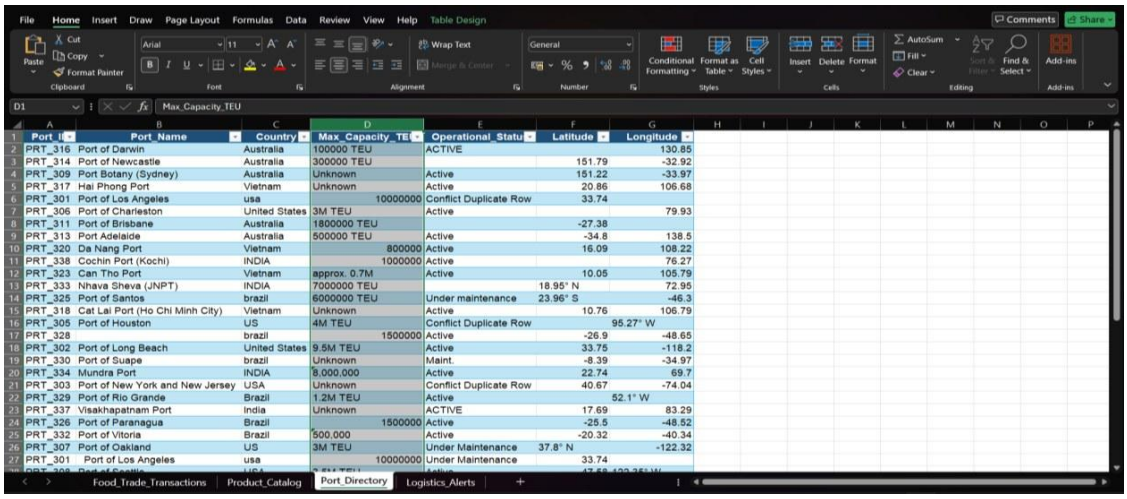
3. **COUNTRY**: the PROPER() function was applied.



Port_ID	Port_Name	Country	Max_Capacity_TEU	Operational_Status	Latitude	Longitude
PRT_316	Port of Darwin	Australia	100000 TEU	ACTIVE		130.85
PRT_314	Port of Newcastle	Australia	300000 TEU		151.79	-32.92
PRT_309	Port Botany (Sydney)	Australia	Unknown	Active	151.22	-33.97
PRT_317	Hai Phong Port	Vietnam	Unknown	Active	20.86	106.68
PRT_301	Port of Los Angeles	USA	10000000	Conflict Duplicate Row	33.74	
PRT_306	Port of Charleston	United States	3M TEU	Active		79.93
PRT_311	Port of Brisbane	Australia	1800000 TEU		-27.38	
PRT_313	Port Adelaide	Australia	500000 TEU	Active	-34.8	138.5
PRT_320	Da Nang Port	Vietnam	800000	Active	16.09	108.22
PRT_338	Cochin Port (Kochi)	INDIA	1000000	Active		76.27
PRT_323	Can Tho Port	Vietnam	approx. 0.7M	Active	10.05	105.79
PRT_333	Nhava Sheva (JNPT)	INDIA	7000000 TEU		18.95° N	72.95
PRT_325	Port of Santos	brazil	6000000 TEU	Under maintenance	23.96° S	-46.3
PRT_318	Cat Lai Port (Ho Chi Minh City)	Vietnam	Unknown	Active	10.76	106.79
PRT_305	Port of Houston	USA	4M TEU	Conflict Duplicate Row		95.27° W
PRT_328		brazil	1500000	Active	-26.9	-48.65
PRT_302	Port of Long Beach	United States	9.5M TEU	Active	33.75	-118.2
PRT_330	Port of Suape	brazil	Unknown	Maint.	-8.39	-34.97
PRT_334	Mundra Port	INDIA	8,000,000	Active	22.74	69.7
PRT_303	Port of New York and New Jersey	USA	Unknown	Conflict Duplicate Row	40.67	-74.04
PRT_329	Port of Rio Grande	Brazil	1.2M TEU	Active		52.1° W
PRT_337	Visakhapatnam Port	India	Unknown	ACTIVE	17.69	83.29
PRT_326	Port of Paranagua	Brazil	1500000	Active	-25.5	-48.52
PRT_332	Port of Vitoria	Brazil	500,000	Active	-20.32	-40.34
PRT_307	Port of Oakland	Australia	3M TEU	Under Maintenance	37.8° N	-122.32
PRT_301	Port of Los Angeles	USA	10000000	Under Maintenance	33.74	

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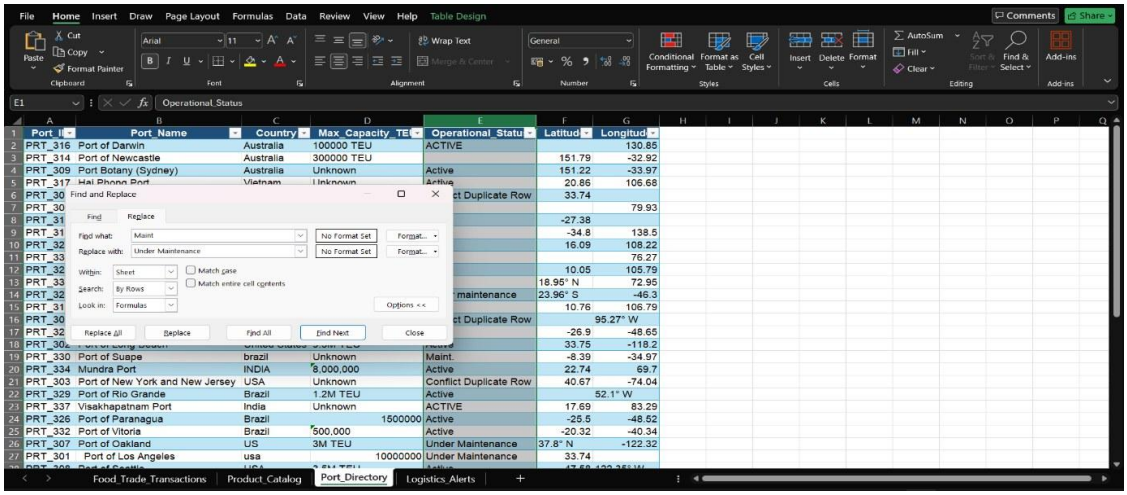
4. **MAX_CAPACITY_TEU**: data inconsistencies were corrected, and missing values were filled by looking up each port by its PORT_NAME using ChatGPT.

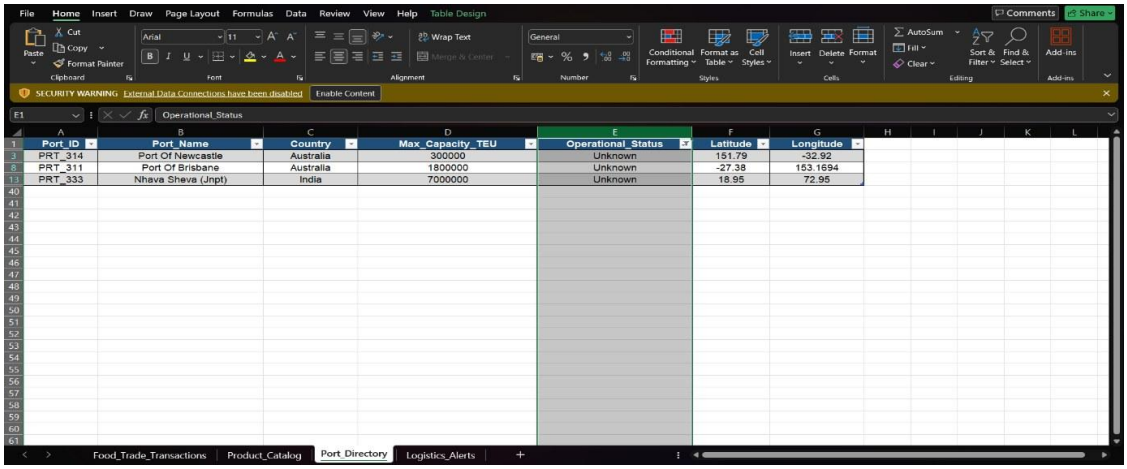


Port_ID	Port_Name	Country	Max_Capacity_TEU	Operational_Statu	Latitude	Longitude
PRT_316	Port of Darwin	Australia	100000 TEU	ACTIVE		130.85
PRT_314	Port of Newcastle	Australia	300000 TEU		151.79	-32.92
PRT_309	Port Botany (Sydney)	Australia	Unknown	Active	151.22	-33.97
PRT_317	Hai Phong Port	Vietnam	Unknown	Active	20.86	106.68
PRT_301	Port of Los Angeles	usa	10000000	Conflict Duplicate Row	33.74	
PRT_306	Port of Charleston	United States	3M TEU	Active		79.93
PRT_311	Port of Brisbane	Australia	1800000 TEU		-27.38	
PRT_313	Port Adelaide	Australia	500000 TEU	Active	-34.8	138.5
PRT_320	Da Nang Port	Vietnam	800000	Active	16.09	108.22
PRT_338	Cochin Port (Kochi)	INDIA	1000000	Active		76.27
PRT_323	Can Tho Port	Vietnam	approx. 0.7M	Active	10.05	105.79
PRT_333	Nhava Sheva (JNPT)	INDIA	7000000 TEU		18.95° N	72.95
PRT_325	Port of Santos	brazil	6000000 TEU	Under maintenance	23.96° S	-46.3
PRT_318	Cat Lai Port (Ho Chi Minh City)	Vietnam	Unknown	Active	10.76	106.79
PRT_305	Port of Houston	US	4M TEU	Conflict Duplicate Row		95.27° W
PRT_328		brazil	1500000	Active	-26.9	-48.65
PRT_302	Port of Long Beach	United States	9.5M TEU	Active	33.75	-118.2
PRT_330	Port of Suape	brazil	Unknown	Maint.	-8.39	-34.97
PRT_334	Mundra Port	INDIA	8,000,000	Active	22.74	69.7
PRT_303	Port of New York and New Jersey	USA	Unknown	Conflict Duplicate Row	40.67	-74.04
PRT_329	Port of Rio Grande	Brazil	1.2M TEU	Active		52.1° W
PRT_337	Visakhapatnam Port	India	Unknown	ACTIVE	17.69	83.29
PRT_326	Port of Paranagua	Brazil	1500000	Active	-25.5	-48.52
PRT_332	Port of Vitoria	Brazil	500,000	Active	-20.32	-40.34
PRT_307	Port of Oakland	US	3M TEU	Under Maintenance	37.8° N	-122.32
PRT_301	Port of Los Angeles	usa	10000000	Under Maintenance	33.74	

Food_Trade_Transactions | Product_Catalog | Port_Directory | Logistics_Alerts

5. **OPERATIONAL_STATUS**: “Maint” was replaced with “Under Maintenance” using Find and Replace, and missing values were filled with Unknown.

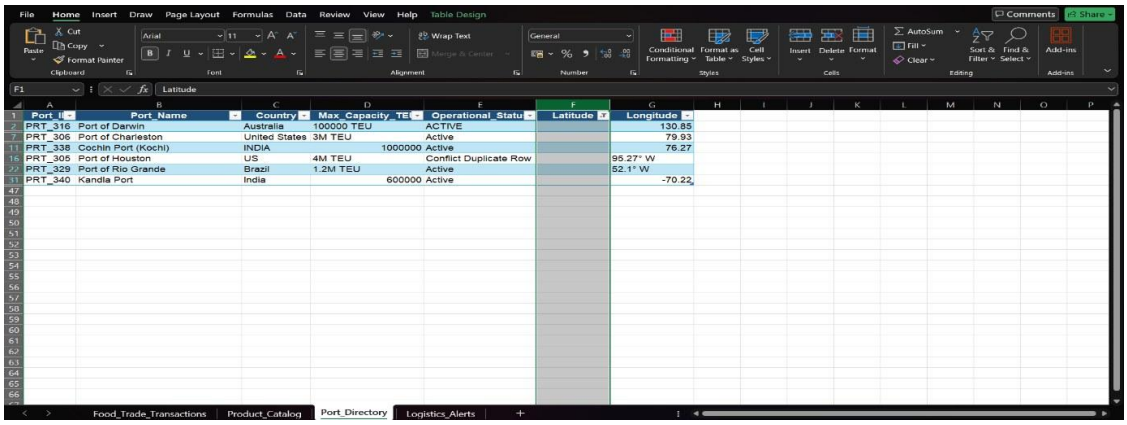




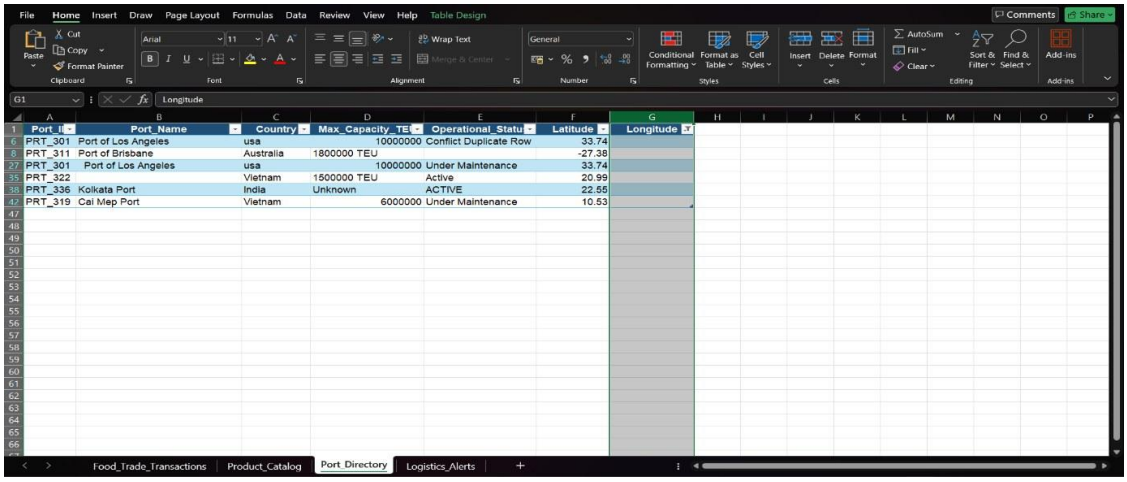
Port_ID	Port_Name	Country	Max_Capacity_TEU	Operational_Status	Latitude	Longitude
PRT_314	Port Of Newcastle	Australia	300000	Unknown	151.79	-32.92
PRT_311	Port Of Brisbane	Australia	1800000	Unknown	-27.38	153.1694
PRT_333	Nhava Sheva (Jnpt)	India	7000000	Unknown	18.95	72.95

Food_Trade_Transactions | Product_Catalog | Port_Directory | Logistics_Alerts

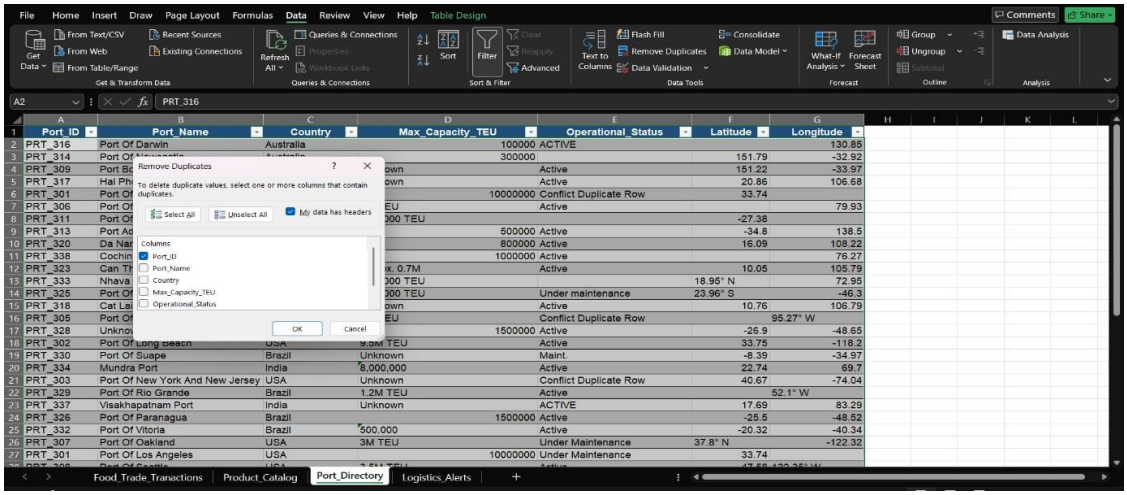
6. **LATITUDE**: missing values were filled by looking up each port by its PORT_NAME using ChatGPT.

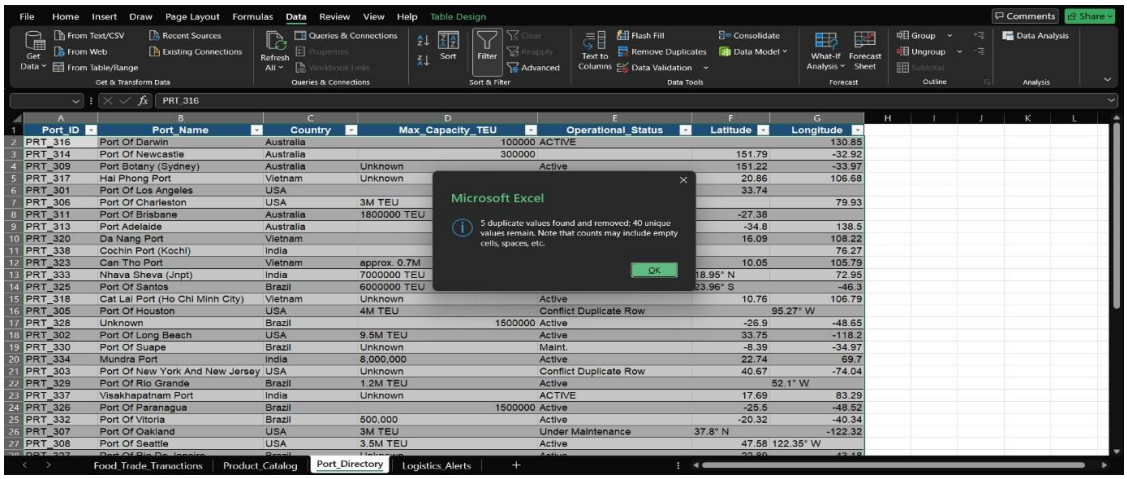


7. **LONGITUDE**: missing values were filled by looking up each port by its PORT_NAME using ChatGPT.



Duplicate removal: 5 duplicate rows were removed using Remove Duplicates on the PORT_ID column.

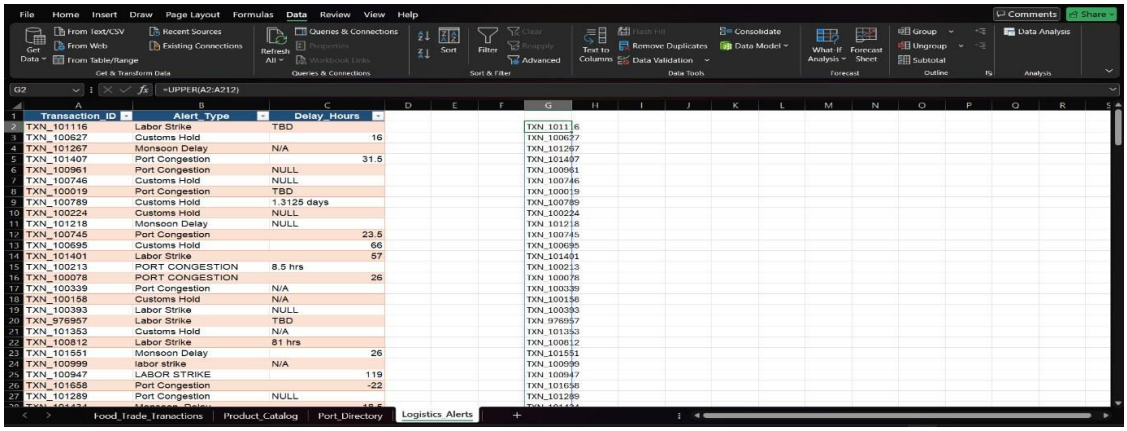




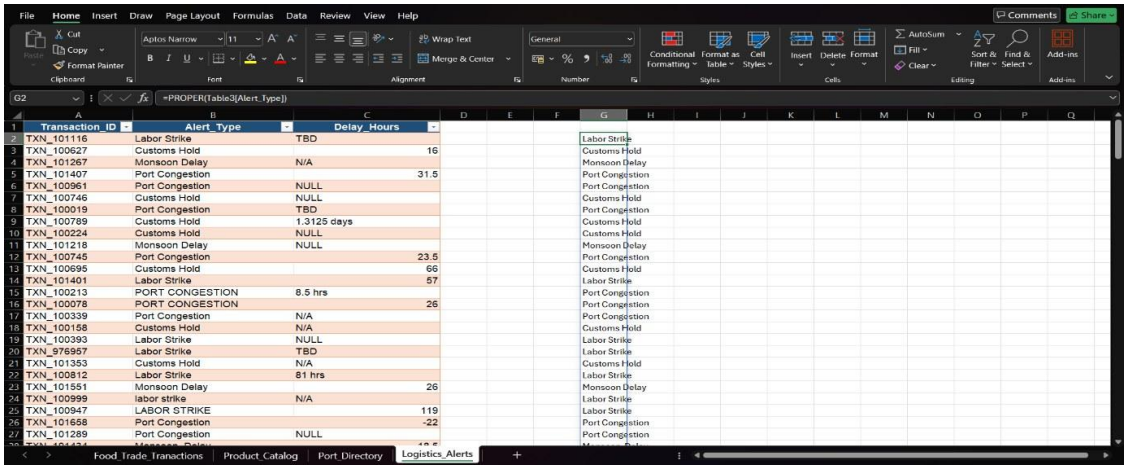
4. Logistics_Alerts

After cleaning and transformation, this table has 193 rows and 3 columns.

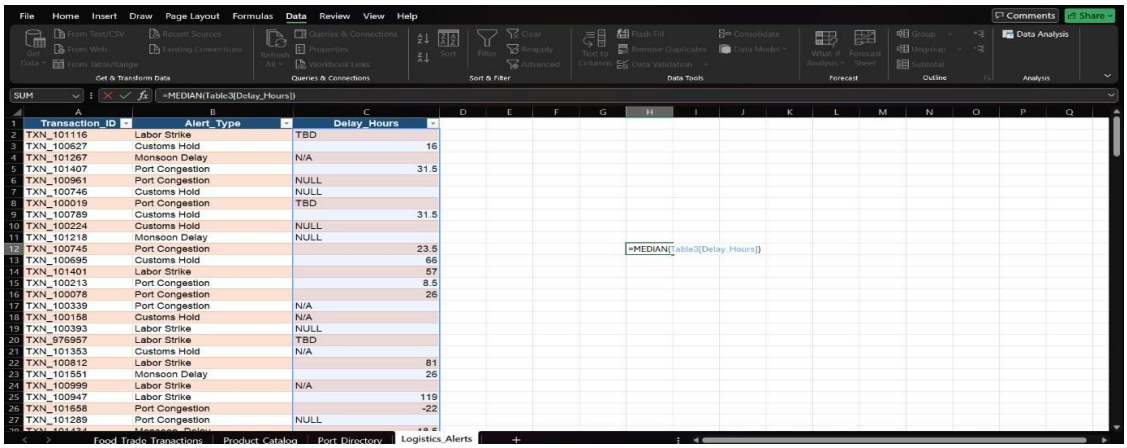
1. **TRANSACTION_ID**: converted to upper case using the UPPER() function (for example, “txn_100116” became “TXN_100116”).



2. **ALERT_TYPE**: the PROPER() function was applied.



3. **DELAY_HOURS**: missing values were filled with the median value (38.25), and data inconsistencies were removed. Values given in days were converted to hours (for example, 1.3125 days × 24 = 31.5 hours).



Duplicate removal: 10 duplicate rows were removed using Remove Duplicates on the TRANSACTION_ID column.

